HSPT Exam Simulator

Full-Length Practice Exam 14

Complete printable test with 298 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Verbal Skills	60	20 minutes
Quantitative Skills	52	35 minutes
Reading Comprehension	62	35 minutes
Mathematics	64	45 minutes
Language	60	30 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 165 minutes. Calculators should be used only when permitted by the rules followed for your HSPT preparation.

Section 1: Verbal Skills

60 questions | Recommended time: 20 minutes

1. Which phrase is closest in meaning to foyer?

A) an upper balcony

B) a storage cellar

C) a rear courtyard

D) an entrance hall

2. What does a smattering of knowledge mean?

A) a complete lack of knowledge

B) a small amount of knowledge

C) a thorough command of a subject

D) an organized method of teaching

3. Which phrase best explains palpable tension between two people?

A) tension that is readily noticeable

B) tension that has completely disappeared

C) tension invented for an audience

D) tension that neither person can detect

4. Which word is most nearly opposite in meaning to waning, as in waning interest?

A) fading

B) increasing

C) declining

D) dwindling

5. A substance described as chemically inert does not readily react. Which word is most nearly opposite in meaning to inert here?

A) reactive

B) inactive

C) unresponsive

D) unchanging

6. Which phrase is most nearly opposite in meaning to offhand in "an offhand judgment made without reflection"?

A) casually offered

B) quickly improvised

C) carefully considered

D) thoughtlessly expressed

7. A ransom is demanded in exchange for release. In a similar relationship, a toll is charged in exchange for ____.

A) ownership

B) employment

C) passage

D) repayment

8. To intercept a message is to stop it before its arrival. To preempt an action is to stop it before its ____.

A) completion after it starts

B) description in a report

C) beginning

D) repetition after it ends

9. A concession relaxes a demand. In a similar relationship, an exception relaxes a ____ in a particular case.

A) rumor

B) rule

C) memory

D) prediction

10. Which word names a final ending rather than a temporary interruption?

A) hiatus

B) lull

C) respite

D) finale

11. Which pair names actions that both normally create an obligation to return what is transferred, unlike giving a gift?

A) lending; borrowing

B) donating; receiving a gift

C) discarding; finding

D) displaying; watching

12. Which word has different pronunciations in its two stated meanings, while retaining exactly the same spelling?

A) bark: a tree covering; a dog's cry

B) match: a contest; a fire-lighting stick

C) bat: a flying mammal; a striking implement

D) wind: moving air; to turn a handle

13. A sign says, "A fine applies for parking here." Someone reasons, "Fine means excellent, so the sign says parking here is excellent." What is the error?

A) The reasoning changes the meaning of the same word between statements.

B) The reasoning compares two different parking locations.

C) The reasoning shows that the sign contains no rule.

D) The reasoning assumes the sign was written yesterday.

14. A claim about an unknown mechanism has not been disproved. A writer concludes that the claim must therefore be true. What is the main weakness in that conclusion?

A) An unknown mechanism can never be investigated by any method.

B) The absence of disproof alone does not establish that the claim is true.

C) A claim without disproof must be false rather than true.

D) The writer has demonstrated that every competing claim is false.

15. A lamp lights exactly when its cell and bulb both work. Exactly one of those two parts is faulty. A working replacement bulb is installed, but the lamp remains dark. What follows?

A) The original bulb must have been faulty.

B) Both original parts must have been faulty.

C) The original cell is faulty.

D) The working replacement bulb proves the cell works.

16. Which phrase best explains showing gumption when faced with a difficult task?

A) waiting for someone else to begin

B) displaying confusion about the instructions

C) showing initiative and determination

D) avoiding every possible difficulty

17. Which phrase is closest in meaning to a foible in someone's character?

A) an outstanding achievement

B) a formal duty

C) an unbreakable promise

D) a minor personal weakness

18. Which phrase best explains the salient points of an argument?

A) the points deliberately omitted from discussion

B) the points that stand out as most important

C) the points supported only by numerical calculations

D) the points placed last regardless of importance

19. Which word is most nearly opposite in meaning to jovial, describing a person's mood?

A) merry

B) cheerful

C) jolly

D) gloomy

20. Which phrase is most nearly opposite in meaning to susceptible to damage?

A) resistant to damage

B) easily harmed

C) likely to be affected

D) readily weakened

21. Which phrase is most nearly opposite in meaning to unilateral, as in a unilateral decision by one of the parties?

A) made jointly by the parties

B) made by one party alone

C) announced well in advance

D) made without telling the other party

22. A boycott involves refusing to buy in order to exert pressure. In a similar relationship, a strike involves refusing to ____.

A) remember

B) work

C) travel for pleasure

D) make a prediction

23. A glossary directs a reader to a term's meaning. In a similar relationship, an index directs a reader to a topic's ____ within a book.

A) pronunciation

B) location

C) truthfulness

D) language of origin

24. An adjective can modify a noun. In a similar grammatical relationship, an adverb can modify a ____.

A) letter of the alphabet

B) punctuation mark

C) syllable

D) verb

25. Which word names an unintended occurrence rather than a planned course of action?

A) strategy

B) mishap

C) policy

D) scheme

26. Which pair uses the same word first as a noun and then as a verb with a related meaning?

A) a heavy box; a heavy crate

B) an early flight; an early train

C) slowly leave; slowly return

D) a soft brush; brush away the dust

27. Which pair contains two words whose apparently negative forms express exceptionally great value?

A) unfair; unjust

B) unbroken; unseen

C) priceless; invaluable

D) unwanted; unwelcome

28. A reviewer says, "I will approve the design if its labels are readable." The labels are made readable. The reviewer then refuses approval solely because the border is not gold, a new requirement. What changed?

A) An additional condition was introduced after the announced condition was met.

B) The design was shown to have unreadable labels.

C) The original statement required a gold border and omitted readability.

D) The new requirement proves that readable labels are impossible.

29. A proposal allows quiet conversation only during the interval between acts. A critic responds, "The proposal demands talking throughout every performance." What is wrong with the response?

A) It accurately preserves the proposal's restriction to the interval.

B) It shows that an interval cannot occur between two acts.

C) It replaces a limited proposal with a much broader position that was not proposed.

D) It establishes that everyone favors the original proposal.

30. Rae must write the exact words printed on a card, rather than carry out the card's instruction. The card reads "Do not write anything." Which response completes Rae's assigned task?

A) I cannot copy this instruction.

B) This card has no words.

C) Do not write anything.

D) I will leave the response blank.

31. In the phrase "a penchant for intricate puzzles," PENCHANT most nearly means:

A) a formal obligation

B) a brief doubt

C) a deliberate insult

D) a strong liking

32. Lena had a SCRUPLE about accepting credit for someone else's work. SCRUPLE most nearly means:

A) an ethical reservation

B) physical exhaustion

C) a desire for applause

D) certainty of a reward

33. The dancer entered with PANACHE. PANACHE most nearly means:

A) a confident, stylish manner

B) careless indifference

C) shy reluctance

D) mechanical accuracy

34. The NADIR is the lowest point. Which phrase is its opposite?

A) the highest point

B) the lowest level

C) a downward trend

D) a temporary rest

35. Which phrase is most nearly opposite in meaning to GERMANE TO THE DISCUSSION?

A) closely connected to the issue

B) directly applicable to the subject

C) pertinent to the argument

D) unrelated to the topic

36. Which description is most nearly opposite in meaning to DOGMATIC?

A) insistent that one's own views cannot be questioned

B) unyielding about an asserted principle

C) willing to reconsider firmly held beliefs

D) certain that disagreement must be mistaken

37. Forced perspective can make objects appear to have different SIZES. In a comparable illusion, ventriloquism makes a voice appear to come from a different:

A) language

B) loudness

C) location

D) duration

38. A sieve separates mixed particles according to SIZE. A prism separates white light according to:

A) weight

B) color

C) temperature

D) age

39. A narrow doorway limits the WIDTH of an object that can pass through it. A low ceiling similarly limits the object's:

A) weight

B) color

C) length

D) height

40. Three words describe continuing to occupy a place. Which word instead names the act of crossing into it?

A) residence

B) entry

C) stay

D) occupancy

41. In ordinary English, which pair consists of two tools whose names normally take plural forms, even when each name refers to one instrument?

A) gloves; socks

B) tongs; spoon

C) shoes; boots

D) scissors; tweezers

42. Start with any row of four distinct letters. Which pair contains two instructions that EACH restore the original row when that same instruction is performed twice?

A) Alphabetize the row; reverse the entire row.

B) Move the first letter to the end; exchange its first and last letters.

C) Reverse the entire row; exchange its first and last letters.

D) Duplicate the first letter; reverse the entire row.

43. A speaker argues, "This practice has been followed for generations, so it cannot be improved." Which response identifies the flaw?

A) The length of time a practice has existed does not establish that it is the best possible practice.

B) Every newer practice is necessarily better than an older one.

C) A practice followed for generations can never produce good results.

D) The speaker has proved that no one has ever questioned the practice.

44. A finite crew has more than half its members able to row and more than half able to sail. Which statement must be true?

A) Every crew member can both row and sail.

B) The rowing and sailing groups have no members in common.

C) At least one crew member can both row and sail.

D) Exactly one crew member has either skill.

45. A card was formerly accepted exactly when it was both blue and square. The revised rule accepts a card when it is blue or square or both. Which statement must be true?

A) Every formerly rejected card is now accepted.

B) Every card accepted under the former rule is accepted under the revised rule.

C) Only blue cards that are not square are now accepted.

D) Cards that are neither blue nor square are now accepted.

46. If a visitor DAWDLES on the way to the exit, the visitor:

A) leaves a useful object behind

B) asks for a different destination

C) takes unnecessary time moving along

D) travels by the shortest route

47. Calling the guest by the wrong name was an unfortunate GAFFE. GAFFE most nearly means:

A) a private agreement

B) an unexpected reward

C) an embarrassing social mistake

D) a deliberate compliment

48. A supervisor was accused of NEPOTISM after appointing less-qualified relatives to several jobs. NEPOTISM most nearly means:

A) favoring family members in appointments

B) choosing applicants by lottery

C) requiring unusually long training

D) promoting workers solely for their achievements

49. LAX enforcement allows rules to be ignored without much concern. Which description is most nearly opposite?

A) casual enforcement

B) strict enforcement

C) inconsistent enforcement

D) lenient enforcement

50. Which phrase is most nearly opposite in meaning to TAWDRY, describing decoration that is showy but cheap-looking?

A) gaudily flashy

B) tastefully elegant

C) cheaply glittering

D) crudely ornamental

51. Which description is most nearly opposite to FINANCIALLY SOLVENT?

A) able to meet financial obligations

B) capable of settling bills

C) possessing enough funds to cover liabilities

D) unable to pay debts

52. A contour line on a map joins locations with equal ELEVATION. In a similar relationship, an isobar on a weather map joins locations with equal air:

A) temperature

B) pressure

C) humidity

D) direction

53. A space in printed text reserves room without a letter. A rest in written music similarly reserves time without:

A) a written symbol

B) a measured duration

C) a sounded note

D) a place in the score

54. A keystroke counter reports the total number of keys pressed, while a typing-rate display reports keys pressed per minute. In the same relationship, a total-volume counter reports accumulated water volume, while a flow-rate meter reports:

A) the container's entire capacity

B) the water's temperature

C) the total distance to the water source

D) water volume passing per unit of time

55. Three expressions guarantee that at least one item exists. Which expression does NOT guarantee this?

A) some items

B) at least one item

C) at most one item

D) not zero items

56. Which item differs from the others because its usual purpose is to respond after a dangerous event, rather than serve as a barrier that helps prevent harm?

A) a floodwall

B) a rescue boat

C) a guardrail

D) a firebreak

57. Which pair consists of words whose plural forms can acquire distinct conventional meanings, rather than merely name several instances of the singular thing?

A) window/windows; river/rivers

B) mouse/mice; tooth/teeth

C) goose/geese; child/children

D) custom/customs; manner/manners

58. A commentator argues, "Most viewers believe the forecast, so the forecast must be accurate." What is missing from this reasoning?

A) Evidence connecting the forecast to the conditions it predicts.

B) Proof that every popular belief is necessarily false.

C) An explanation of why a forecast must please every viewer.

D) A requirement that only unpopular forecasts can be accurate.

59. A researcher studies only old wooden vessels still afloat and concludes, "Wooden vessels almost never sink." Which missing evidence most directly weakens that conclusion?

A) Records of wooden vessels that sank and therefore could not enter the study.

B) The present paint colors of the vessels still afloat.

C) The researcher's preferred kind of vessel.

D) The names given to the surviving vessels when they were built.

60. Each strand of a thick rope can be snapped easily by hand. Someone concludes that the complete rope must also snap easily by hand. What has the person assumed without justification?

A) That a rope contains more than one strand.

B) That the separate strands can be examined.

C) That the rope and its strands are made of material.

D) That a property of each separate component must also hold for the assembled whole.

Section 2: Quantitative Skills

52 questions | Recommended time: 35 minutes

61. To obtain each next term, move the first digit to the end without changing any digits. What comes next?
481, 814, 148, 481, ____

A) 148

B) 814

C) 418

D) 841

62. A sequence has a(1) = 1. For positive integers k, a(2k) = a(k) and a(2k + 1) = a(k) + a(k + 1). The first twelve terms are shown. What is a(13)?
1, 1, 2, 1, 3, 2, 3, 1, 4, 3, 5, 2, ____

A) 5

B) 3

C) 7

D) 8

63. Term n counts strings made from n opening and n closing parentheses that are balanced: no initial portion has more closings than openings. The first three counts are 1, 2, and 5. What is the fourth count?

A) 8

B) 12

C) 16

D) 14

64. Start with 2. Each next term is the smallest positive integer not previously used that shares a factor greater than 1 with the preceding term. What comes next?
2, 4, 6, 3, 9, 12, 8, ____

A) 5

B) 10

C) 7

D) 14

65. A three-digit integer is divisible by 11 if its hundreds digit minus its tens digit plus its units digit is a multiple of 11. Which digit replaces d in 6d2 to make it divisible by 11?

A) 0

B) 4

C) 8

D) 6

66. Write the digits of any three-digit positive integer twice in a row; for example, 247 becomes 247247. Dividing the resulting six-digit integer by the original integer always gives what quotient?

A) 1,001

B) 101

C) 1,000

D) 1,010

67. How many distinct numerical values appear in this list?
1/8, 0.125, 12.5%, 0.0125, 1.25%

A) 1

B) 2

C) 4

D) 5

68. Each arrangement uses all three letters exactly once. Repeated copies of the same letter are indistinguishable.
Quantity A: The number of distinct arrangements of AAB
Quantity B: The number of distinct arrangements of ABB

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

69. The positive integer n is even.
Quantity A: The number of different prime factors of n
Quantity B: The number of factors of 2 in the prime factorization of n

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

70. Quantity A: The number of positive real solutions of $x^2 = 9$
Quantity B: The number of real solutions of $x^2 = 9$ without a sign restriction

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

71. One side of a nondegenerate triangle is extended beyond a vertex, forming an exterior angle.
Quantity A: That exterior angle
Quantity B: Either one of the two nonadjacent interior angles

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

72. Two solid prisms have congruent triangular bases and the same perpendicular distance between their base planes. One prism is right; the other leans sideways.
Quantity A: The volume of the right prism
Quantity B: The volume of the leaning prism

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

73. Two circles of different radii and with different centers intersect at exactly two points.
Quantity A: The length of their common chord
Quantity B: The diameter of the smaller circle

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

74. List positive integers that are NOT divisible by any perfect square greater than 1. What comes next?
1, 2, 3, 5, 6, 7, 10, 11, 13, 14, 15, ____

A) 16

B) 18

C) 17

D) 20

75. Start with 2. Each later term is 1 more than the product of ALL earlier terms. What comes next?
2, 3, 7, 43, ____

A) 1,807

B) 301

C) 1,806

D) 1,849

76. Begin with the numbers 2, 5, 6, 13, and 20. Repeatedly remove the two smallest numbers and insert their sum. The recorded sums are shown. What is the next recorded sum?
7, 13, 26, ____

A) 33

B) 46

C) 39

D) 52

77. If a:b = 3:4 and b:c = 6:5, what is a:c?

A) 1:2

B) 3:5

C) 9:20

D) 9:10

78. Positive integers a and b are both greater than 1 and satisfy $a^2 = b^3$. What is the least possible value of a + b?

A) 9

B) 17

C) 12

D) 18

79. An integer is divided by 5, leaving remainder 4. Its quotient is then divided by 3, leaving remainder 2. What remainder does the original integer leave when divided by 15?

A) 14

B) 2

C) 6

D) 9

80. Which listed number is the sum of TWO DIFFERENT positive perfect squares?

A) 8

B) 10

C) 9

D) 16

81. A list contains seven real numbers. A second list repeats each entry of the first list three times.
Quantity A: The median of the first list
Quantity B: The median of the second list

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

82. For positive numbers, R(x) rounds x to the nearest integer, with exact halves rounded upward. The positive numbers a and b are otherwise unspecified.
Quantity A: R(a) + R(b)
Quantity B: R(a + b)

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

83. A container holds at least one red token and at least one blue token, with no other colors. Tokens are equally likely to be drawn.
Quantity A: The probability of red before any removal
Quantity B: The probability of red after one blue token is removed

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

84. A cube is sliced by a plane through two vertical edges that are diagonally opposite when viewed from above.
Quantity A: The area of the rectangular slice
Quantity B: The area of an original square face

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

85. In triangle ABC, AB = AC, and M is the midpoint of BC.
Quantity A: Angle BAM
Quantity B: Angle MAC

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

86. Triangles ABC and ABD have the same area and common base AB. C and D lie on opposite sides of the line through AB.
Quantity A: The length CD
Quantity B: Twice the perpendicular height from C to line AB

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

87. For each term, separate its units digit from the number formed by all preceding digits. Double that preceding number and add the units digit to obtain the next term.
392, 80, 16, ___

A) 8

B) 6

C) 16

D) 32

88. All four-digit numbers using 1, 3, 6, and 8 exactly once each are listed in increasing order. What comes next?
1368, 1386, 1638, 1683, 1836, ___

A) 1638

B) 1836

C) 3168

D) 1863

89. A sequence of four-place binary numbers follows alternating moves R, L, R, L, and so on. A flip changes 0 to 1 or 1 to 0.
R: Flip the rightmost digit.
L: Flip the digit immediately to the left of the rightmost 1.
0000, 0001, 0011, 0010, 0110, 0111, 0101, 0100, ___
Which number comes next?

A) 0101

B) 1100

C) 1000

D) 1101

90. Available box capacities are 1, 2, 4, 8, 16, 32, and so on, doubling each time. Term n is the smallest available capacity that can hold n items.
The sequence begins 1, 2, 4, 4, 8, 8, 8, 8. What is its seventeenth term?

A) 16

B) 17

C) 32

D) 64

91. What is the value of $(4 \times 10^{\square}) \div (8 \times 10^{2})$?

A) 50

B) 5000

C) 0.5

D) 500

92. A loom pattern has light and dark threads in the ratio 3:5. Four dark threads are replaced one for one by four light threads, making the two counts equal. How many threads were in the original pattern?

A) 32

B) 16

C) 24

D) 40

93. Which expression is equal to 1 ÷ (√5 − 2)?

A) √5 − 2

B) √5 + 2

C) 2 − √5

D) (√5 + 2) ÷ 9

94. For events E and F, P(E) > P(F).
Quantity A: The probability that E does not occur
Quantity B: The probability that F does not occur

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

95. A list contains positive numbers. A second list is formed by multiplying every number in the first list by 4. Range means greatest value minus least value.
Quantity A: The first list's range divided by its mean
Quantity B: The second list's range divided by its mean

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

96. The real numbers a and b are distinct.
Quantity A: The larger of a and b
Quantity B: (3a + 7b) ÷ 10

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

97. A 7-by-12 rectangular sheet is folded exactly in half along a line parallel to one pair of its sides. The pair is not specified.
Quantity A: The perimeter of the resulting folded rectangle
Quantity B: 28 units

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

98. A right triangle has positive integer side lengths and a perimeter of 60 units.
Quantity A: The length of its longer leg
Quantity B: 22 units

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

99. Two straight lines intersect.
Quantity A: The measure of one angle at the intersection
Quantity B: The measure of the vertically opposite angle

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

100. Begin with a remainder of 19/20. Repeatedly subtract the largest unit fraction 1/k that does not exceed the current remainder, recording its denominator k. The first recorded denominators are 2, 3, and 9. What denominator is recorded next?

A) 60

B) 9

C) 120

D) 180

101. To obtain each next term, split the six digits into two blocks of three. Interleave the blocks, taking a digit from the SECOND block first each time.
123456, 415263, 246135, ___

A) 214635

B) 135246

C) 123456

D) 124365

102. Replace every 1 with 12, every 2 with 3, and every 3 with 1. Make all replacements simultaneously, without replacing newly written digits again in that step.
1, 12, 123, 1231, 123112, ___

A) 123112123

B) 12311212

C) 123123112

D) 1231121231

103. Which real number, when cubed, equals −216?

A) 6

B) −6

C) −72

D) −36

104. Write an integer as exactly three digits, allowing initial zeros. Replace each digit d by 9 − d to make a second three-digit string. What is always the sum of the two represented values?

A) 900

B) 1000

C) 999

D) 909

105. In base b, the rightmost place counts units and the next place counts groups of b. The numeral 21 in base b represents the decimal number 15. What is b?

A) 6

B) 15

C) 14

D) 7

106. A positive fraction p/q lies strictly between 2/5 and 3/7, with positive integers p and q. What is the smallest possible value of q?

A) 12

B) 5

C) 11

D) 17

107. A raffle has W winning tickets and L losing tickets, where W and L are positive. Each ticket is equally likely to be drawn.
Quantity A: The odds in favor of winning, W/L
Quantity B: The probability of winning, W/(W + L)

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

108. The real number x satisfies 0 < x < 1.
Quantity A: x^2
Quantity B: x

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

109. The integer n is at least 2. The notation n! means 1 × 2 × ... × n.
Quantity A: n!
Quantity B: (n/2) raised to the power n

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

110. Two parallel chords lie in the same circle of radius 6 units. The first chord is 2 units from the center; the second is 5 units from the center.
Quantity A: The length of the first chord
Quantity B: The length of the second chord

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

111. All four vertices of a quadrilateral lie on one circle.
Quantity A: The sum of a pair of opposite interior angles
Quantity B: 200°

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

112. Two distinct straight rails are parallel. At two different locations, their separation is measured along a perpendicular from one rail to the other.
Quantity A: The first perpendicular separation
Quantity B: The second perpendicular separation

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

Section 3: Reading Comprehension

62 questions | Recommended time: 35 minutes

Literary Passage: To Be Opened This Year

At eleven, Lina had written to the person she expected to be at fourteen. Now that person sat on the rug, holding an envelope addressed in enormous, confident letters.

The letter began with questions. Could she speak another language without stopping? Had she decided what she would become? Did she finally know how to answer every difficult question without blushing? Beneath the questions, her younger self had written, "These are the only things worth knowing by then."

Lina read the sentence twice. She could ask for directions in another language, but she still needed people to speak slowly. Her career plans changed regularly. Blushing remained beyond her control. The letter seemed to have charged her with arriving late to her own life.

She turned it over. There was nothing on the back: no question about finding the bus that took Grandfather to his appointments, or about keeping a friendship after Kian moved away. She had learned those things without first putting them on a list.

Lina reached for a pen. She did not cross out the questions. They were still questions she cared about. Instead, she drew one careful line through "the only things."

Then she put a blank sheet beside the letter. For a while she rested her hand on it without writing. This time she wanted to leave some room for a person she had not met yet.

113. What does Lina cross out in her earlier letter?

A) the words "the only things"

B) the questions about language and work

C) the address on the envelope

D) everything written on the front

114. In "charged her with arriving late to her own life," charged most nearly means ____.

A) paid

B) supplied with energy

C) gave responsibility for organizing

D) accused

115. Why are Grandfather's bus and Lina's friendship with Kian important to her response?

A) They establish that her original questions have become impossible to answer.

B) They prove that her younger self already knew all her future responsibilities.

C) They show that completing a written list prevents other kinds of learning.

D) They show that meaningful learning occurred outside her earlier predictions.

116. What effect does the description "enormous, confident letters" create at the beginning?

A) It contrasts her younger certainty with her older, more questioning response.

B) It establishes that another person secretly wrote the letter.

C) It makes the unopened envelope seem physically dangerous.

D) It shows that Lina has forgotten how to read her own handwriting.

117. The final reference to "a person she had not met yet" primarily presents Lina's future self as ____.

A) a stranger who will take her place in the family

B) someone whose growth cannot be fully prescribed by her present expectations

C) a critic who must approve every question before she writes

D) someone whose goals will certainly be identical to her childhood goals

Expository Passage: A Shape Without the Shell

A fossil need not contain any of the original material of the organism whose shape it records. One way this can happen begins with a shell becoming buried in sediment. The sediment presses against the shell's surface and may eventually harden into rock. Later, water can dissolve the shell while leaving the surrounding rock in place. The resulting cavity is called an external mold when it preserves an impression of the shell's outer surface. The mold is a hollow form: where the shell once bulged outward, the surrounding rock contains an inward space. It records a boundary without containing the object that made that boundary.

Sometimes new sediment or minerals fill this space. If the filling hardens, it can form a cast with the shell's outward shape. The cast is solid where the mold was hollow, but its material may be entirely different from the vanished shell. Making a cast changes the form in which the evidence is preserved; it does not bring the original shell back.

These traces also have limits. A mold that preserves only the outer surface may record ridges and curves while revealing nothing about the animal's original color. Recognizing what sort of trace remains helps a reader of the rocks distinguish a preserved feature from information the trace never retained.

118. What event leaves the cavity described in the passage?

A) The buried shell dissolves while the surrounding rock remains.

B) The original animal grows a second shell inside the first.

C) The surrounding rock dissolves while the complete shell remains.

D) New sediment turns the original shell into a living animal.

119. How can a cast form in the process described?

A) The hollow mold must remain completely empty forever.

B) The original animal returns and grows into the cavity.

C) The rock surrounding the mold is compressed into an unchanged original shell.

D) New sediment or minerals fill the mold and harden.

120. Which idea best unites the shell's disappearance and the formation of a mold or cast?

A) All fossils consist entirely of the original materials of once-living organisms.

B) Molds and casts can preserve aspects of a shell without preserving the original shell itself.

C) A shell can leave a useful fossil only if its original color remains visible.

D) Molds and casts are two names for an identical hollow structure.

121. A hollow external mold shows several ridges from a shell, but no cast has formed. Which conclusion is best supported by the passage?

A) The original color can be recovered from the ridge pattern alone.

B) The cavity must contain a complete shell made of its original material.

C) The ridge pattern may be studied even though the shell itself is gone.

D) No evidence of the outer surface remains until the cavity is filled.

122. As used in "The resulting cavity," cavity most nearly means ____.

A) a living creature

B) a hollow space

C) a solid filling

D) a patch of color

Reflective Passage: A Good Move That Loses

A player studies a board and chooses a move. An unlikely draw then turns that promising position into a loss. At the next table, another player overlooks an obvious danger, makes a careless move, and wins when a fortunate draw removes the danger. Which player made the better decision?

The final scores alone cannot answer that question. A decision is made with information available before its result is known. Its quality depends on how reasonably that information was used, not simply on whether later events were favorable. If the first player considered the relevant possibilities and the second ignored them, a loss and a win need not reverse that assessment.

This distinction does not excuse every mistake. A player cannot defend a poor move merely by saying that anything might happen. An avoidable risk visible at the time still counts against the decision. Nor does the distinction make results useless: a long record of outcomes may reveal a weakness that one isolated game cannot establish.

What the single fortunate draw cannot warrant is a claim that carelessness was wise. When reviewing a choice, it helps to place an imaginary screen between the moment of decision and the moment of revelation. First ask what could reasonably have been known on the near side of that screen. Only then consider what happened beyond it.

123. What removes the danger facing the careless player?

A) a warning from the other player

B) a deliberate revision of the rules

C) an analysis made before the move

D) a fortunate draw

124. Which principle about judging decisions does the whole passage develop?

A) A winning result proves that the reasoning behind a choice was sound.

B) A losing result shows that every available alternative would have been better.

C) A choice should be judged using what was reasonably knowable when it was made, as well as a careful view of later results.

D) A choice should never be evaluated because future events are uncertain.

125. Which criticism of a losing move would the author most likely consider fair?

A) The player failed to know a concealed future draw that no available evidence predicted.

B) The player lost, so the move must have been the worst possible choice.

C) The player ignored a danger clearly visible on the board before choosing.

D) Another player won elsewhere, so the losing player should have copied that player's move.

126. As used in "cannot warrant ... a claim," warrant most nearly means ____.

A) justify

B) repeat aloud

C) postpone

D) make surprising

127. Why does the author include the third paragraph?

A) To withdraw the claim that the final score alone is insufficient

B) To prevent the main distinction from being stretched into excuses for every mistake or rejection of all results

C) To replace the board-game example with instructions for a different game

D) To argue that a long record is always less useful than one isolated win

128. What is the main function of the imaginary screen in the final paragraph?

A) It hides the player's mistakes from anyone reviewing the game.

B) It prevents future draws from changing the actual result.

C) It separates information available at the choice from information revealed afterward.

D) It divides winning players from losing players at separate tables.

Literary Passage: The Name on the Tag

The repair clerk tucked Soraya's boots into a paper bag and reached for a tag. "Sara?" he asked, already lowering his pen.

Soraya had given her name, but she nodded. Close enough was a phrase she often used without saying it. On forms, at counters, when someone called across a room, she accepted names that nearly belonged to her. Correcting them seemed to require a space larger than the little error itself.

The clerk wrote SARA. Behind her, someone set down a parcel. A clock over the shelf gave a dry click.

"Actually," Soraya began, and then stopped. She imagined the parcel being lifted again, its owner sighing. She imagined explaining herself for the rest of the afternoon.

The clerk looked up. "Yes?"

"Soraya. There's an o, and a y."

"Let me start again." He tore off the tag and took another. His pen moved slowly this time, waiting while she spelled. The person behind her was reading a notice about opening hours.

SORAYA. The clerk turned the tag toward her, and she nodded again. This nod felt different from the first.

Outside, she checked the time. Not even a full minute had passed. The street was carrying on with its ordinary business. She had expected to hold everything up; instead, for a few seconds, someone had simply held a place for her name.

129. What does the clerk first write on the tag?

A) SORAYA

B) the opening hours

C) the name of the person behind her

D) SARA

130. Which statement best captures the change at the center of Soraya's experience?

A) She learns that the clerk deliberately misspells every customer's name.

B) She decides that accepting any mistaken name is the quickest way to build friendships.

C) She discovers that asking for her name to be correct can be a small, reasonable request.

D) She discovers that repairing the boots will require an entire afternoon.

131. What does the waiting person's actual behavior suggest about Soraya's imagined sigh?

A) The waiting person secretly demanded that the clerk keep the first tag.

B) The waiting person had already left without being served.

C) The clerk asked the waiting person to correct Soraya's spelling.

D) She had anticipated impatience that the scene does not show.

132. What does the list beginning "On forms, at counters, when someone called across a room" contribute to the characterization?

A) It proves that Soraya has never visited the repair shop before.

B) It presents accepting approximate names as an established habit across several situations.

C) It limits her concern to written forms and excludes spoken encounters.

D) It shows that every person in those settings intentionally wishes to upset her.

133. What does the final contrast between "hold everything up" and "held a place for her name" accomplish?

A) It establishes that the clerk has physically blocked the shop entrance.

B) It proves that Soraya intends to return under a different name.

C) It transforms the idea of holding from causing delay into making room for someone.

D) It changes the scene from the repair shop to a public ceremony.

Expository Passage: An Image Through a Small Hole

A dark box with a small hole on one side can form an image on the inside surface opposite the hole. No lens is necessary for this simple arrangement. In a basic ray model, the explanation depends on light traveling in straight lines and on the small opening limiting which paths enter.

Trace a ray from the top of an object through the hole. Continuing straight across the box, it reaches a lower part of the opposite surface. A ray from the bottom of the object reaches a higher part. In the same way, rays from one side of the object cross toward the other side of the image. The rays cross at the opening; they do not curve there. The image is inverted because of this arrangement of straight paths.

Why not remove most of the box's front and let in much more light? A large opening admits light from each object point along many different paths. Each point then illuminates a broader patch of the opposite surface, and neighboring patches overlap. More light is admitted, but the separate details become less distinct.

The small opening therefore helps preserve a mapping between points on the object and regions of the image. Brightness alone is not the purpose: an image also needs the contributions from different object points to remain sufficiently separated.

134. According to the ray description, light from the top of the object reaches which part of the opposite surface?

A) a higher part

B) a lower part

C) only the outside of the box

D) the exact same point as every other ray

135. Where does the passage say the image forms?

A) on the inside surface opposite the hole

B) on the object's original surface

C) only on a lens placed outside the box

D) on the outside surface surrounding the opening

136. What explanation connects the passage's discussion of image orientation and opening size?

A) A dark box bends every entering ray upward to correct its direction.

B) A larger opening always produces a more detailed image because brightness is all that matters.

C) A lens is necessary to make rays cross at any opening.

D) Restricting straight light paths helps form a separated, inverted image.

137. As used in "More light is admitted," admitted most nearly means ____.

A) allowed to enter

B) confessed to be true

C) proved to be false

D) sent back out

138. If two small, separated openings each allow light through the same front panel, what does the passage's ray model most strongly suggest?

A) Each opening can project its own point mapping, so the resulting images may overlap.

B) Both openings must bend their rays so that only one upright image remains.

C) The box must become incapable of receiving any light at all.

D) The second opening can only make the first opening's projected paths narrower.

Expository Passage: When a River Leaves a Loop

A winding river does not necessarily keep the same channel forever. In a pronounced bend, flowing water can wear away an outer bank while sediment accumulates along the inner bank. As these changes continue, neighboring arms of a large loop may approach each other. The narrow strip of land between them is often called the neck.

During a flood or another period of strong flow, water may break across this neck. The new route connects the neighboring arms directly, bypassing the long loop. Once substantial flow takes the shorter channel, less water moves through the older bend. Deposited sediment can eventually block its entrances, leaving a curved body of water separated from the active river. This is an oxbow lake.

The lake's outline is a trace of the river's earlier route. Its shape does not mean that someone dug a curved basin beside a permanently fixed river. Nor must the separated loop remain an open lake indefinitely. Sediment and plant material may gradually fill it, turning it into a marsh or a partly dry depression.

These stages need not occur at the same pace everywhere. The point is that a river can leave evidence of an earlier course outside its present channel. A landscape that looks still can contain the record of a waterway that has moved.

139. What process does the first paragraph associate with an outer bank of a pronounced bend?

A) permanent protection from all erosion

B) wearing away by flowing water

C) accumulation of all the river's plant material

D) construction of an artificial basin

140. Which statement best unifies the passage's account of the river and the curved lake?

A) A river must stop flowing completely before any lake can form nearby.

B) Every curved lake is an artificial basin dug next to an unchanged river.

C) A changing river can abandon a loop that preserves evidence of its former course.

D) An oxbow lake proves that a river will return to its former route immediately.

141. A marsh forms a horseshoe shape beside the present river. Which interpretation is supported as a possibility by the passage?

A) It may occupy an abandoned loop that later partly filled in.

B) It must have been an open lake at exactly this time last year.

C) It proves that no river ever occupied that curved area.

D) It must connect to the present river at both ends at all times.

142. In the first paragraph, neck refers to ____.

A) the deepest point inside the lake

B) the outer bank along the entire river

C) the narrow strip of land between neighboring arms of the loop

D) the complete original river channel

143. What is the role of the second paragraph in the explanation?

A) It lists unrelated names for every kind of lake.

B) It rejects the possibility that water can cross a narrow strip of land.

C) It describes a permanent lake first and then claims the river never changed.

D) It traces how a shortcut can redirect flow and leave the old bend separated.

144. Which summary preserves the passage's sequence without turning possible changes into guaranteed events?

A) Every river instantly becomes a marsh whenever sediment reaches an inner bank.

B) A bend may narrow, flow may cut across its neck, and the abandoned loop may later become a lake that gradually fills.

C) A lake must first be dug by people before a river can develop a bend beside it.

D) Every flood leaves all river channels exactly where they were and prevents later filling.

Literary Passage: The Shipping Room's Triumph

Director Kelm announced that the shipping room would become a model of orderly work. The clerks first arranged its parcels alphabetically by destination. When they finished, Kelm noticed that short parcels disappeared behind tall ones. He ordered a second arrangement, by height. Now every parcel was visible, although no clerk could quickly find a particular town.

Kelm considered this an advance in readiness. He then observed that brown wrapping beside blue wrapping gave the room an uneven appearance. The clerks began again, grouping parcels by color. Outside, three drivers waited beside their empty wagons.

“Which parcel may I take?” one driver asked.

“That depends on the final arrangement,” Kelm replied. “Moving a parcel now would leave a gap.”

As the afternoon shortened, a clerk suggested sending the nearest deliveries first. Kelm rejected the suggestion because distant and nearby destinations were distributed among several handsome rows. Selecting parcels by distance would disturb them all. He asked instead for a ledger recording each row's exact order.

At closing time, the ledger was complete. The drivers put away their unused harnesses. Kelm walked between the rows, pleased that not a single parcel stood out of place. In his daily report he wrote, “Our readiness is now complete.” Beyond the doors, the road remained empty; within them, every parcel remained ready.

145. According to the passage, how were the parcels arranged first?

A) From shortest to tallest

B) In groups of the same wrapping color

C) From nearest to most distant delivery

D) Alphabetically by destination

146. Which statement best expresses the central point of the shipping-room episode?

A) Perfecting an arrangement can become a substitute for accomplishing the work the arrangement should serve.

B) Neat arrangements eventually allow the room to complete all its deliveries.

C) The drivers are to blame for refusing to disturb the orderly rows.

D) An accurate ledger matters more than deciding how parcels are grouped.

147. Which proposed action would most directly conflict with the reason Kelm gives for keeping the parcels in the room?

A) Letting a driver remove one parcel from an otherwise complete row.

B) Writing a more detailed record of a row's arrangement.

C) Examining the rows without touching any parcels.

D) Keeping the drivers waiting while the ledger is finished.

148. What makes the narrator's phrase "an advance in readiness" humorous in its context?

A) It repeats the drivers' shared judgment that sorting is almost finished.

B) It announces that the clerks have found a quicker route to every town.

C) It suggests that Kelm is openly criticizing his own method.

D) It uses formal language of progress for a change that has made destinations harder to find.

149. How does the final contrast between the empty road and the full shipping room affect Kelm's triumphant report?

A) It reveals that Kelm modestly understates the completed deliveries.

B) It suggests that the ledger has recorded the rows inaccurately.

C) It allows readers to recognize a failure that Kelm presents as a success.

D) It explains that dangerous road conditions have prevented deliveries.

Literary Passage: The Hail on the Tin Roof

I was peeling an orange beside the open window when something white struck the sill and bounced into the room. For a moment I took it for a pebble. Then another struck, and a pale grain melted against my thumb. Across the lane, the farthest roofs began to hiss. The sound approached in a quick, uneven line. Our own tin roof answered with a clatter so sharp that I stopped moving my hands. The orange peel hung from them in a single curling ribbon. On the sill, the little white stones leaped and spun as if the wood had become a hot pan.

The lane vanished behind a shifting veil. I could still see the opposite doorway, but its straight edges wavered through the falling hail. For several breaths the clatter filled the room so completely that I could hardly remember the quiet that had come before it.

Then the blows grew farther apart. The storm's force over our roof was spent, and water began to tick softly from the gutter. Farther down the road, however, the roof of the long shed still rattled fiercely. I watched until that building came clearly into view again.

The orange was still in my hand. I turned it and continued peeling, extending the ribbon by another small curl.

150. What is the narrator doing immediately before the hail first strikes the sill?

A) Closing the opposite doorway

B) Peeling an orange beside an open window

C) Walking toward the long shed

D) Picking pebbles out of the gutter

151. Which summary best captures the passage as a whole?

A) An unexpected storm permanently changes the narrator's approach to ordinary tasks.

B) A brief hailstorm overwhelms an ordinary moment before the narrator resumes a simple activity.

C) A narrator observes a hailstorm mainly to explain the science behind its formation.

D) A brief shower prevents the narrator from finishing a task for the rest of the day.

152. What can reasonably be inferred when the narrator's roof grows quiet but the long shed still rattles?

A) The shed must have suffered more damage than any other building.

B) The storm continues in another place after its strongest effects have passed the narrator's house.

C) The difference in sound must be caused by different roof materials alone.

D) All the hail has stopped throughout the area.

153. In "The storm's force over our roof was spent," SPENT most nearly means:

A) paid for

B) assigned a task

C) used up

D) borrowed for later

154. Why does the passage end by returning to the orange peel?

A) It frames the intense storm as an interruption within an ordinary activity that continues.

B) It turns the storm into a practical lesson about improving the task.

C) It introduces a separate problem that replaces the storm as the main conflict.

D) It shifts into a flashback about events before the narrator opened the window.

Informational Passage: The Gap in a Vacuum Flask

A vacuum flask slows the movement of heat between its contents and its surroundings. Its double walls enclose a narrow space from which nearly all the air has been removed. This evacuated gap is important, but it is only one part of the design.

Heat can move through matter by conduction. It can also be carried by moving liquids or gases, a process called convection. With almost no material in the gap, there is little to conduct heat directly across it and almost no gas available to circulate there. Removing the air therefore blocks two important routes across that space.

Radiation presents a different problem. Energy carried by radiation can cross empty space; it does not need moving air or a continuous material connection. Reflective surfaces facing the gap reduce this transfer. The vacuum and the reflective surfaces serve different purposes, even though both help slow heat exchange.

The opening needs attention too. A well-fitting stopper limits heat movement through the flask's mouth. Supports that hold the walls apart still provide small material connections through which some heat can pass.

Consequently, the design slows heat transfer rather than ending it completely. It can help keep a hot drink warm or a cold drink cool because the barriers resist heat movement in either direction; the flask itself supplies no heating or cooling.

155. Which process does the passage specifically say can carry energy across empty space?

A) Radiation

B) Convection by circulating air

C) Conduction through a continuous material connection

D) Liquid flowing through a stopper

156. What is the main idea of the explanation of the vacuum flask?

A) The vacuum supplies energy that actively heats a drink.

B) A reflective coating makes the stopper unnecessary.

C) One empty gap eliminates every possible movement of heat.

D) Several design features restrict different paths by which heat could enter or leave.

157. Suppose the reflective surfaces were removed while the gap remained evacuated. Which result is best supported by the passage?

A) The gap would immediately become filled with circulating air.

B) The flask would begin producing heat of its own.

C) Radiation across the gap would be less restricted, while gas circulation there would still be prevented.

D) Every route of heat transfer would be blocked more effectively.

158. As used in the first paragraph, EVACUATED describes a gap that has:

A) been filled with hot liquid

B) had its air removed

C) been widened by stretching

D) had its walls painted

159. How does the final paragraph develop the discussion of the flask's walls?

A) It gives a chronological account of the flask's invention.

B) It argues that distinguishing radiation from convection is unnecessary.

C) It attributes the flask's usefulness entirely to the stopper rather than the walls.

D) It adds other heat paths and clarifies why the design cannot stop heat transfer completely.

Literary Passage: When the Door Became a Line

The door was supposed to open at the visitor's third knock. I had checked its latch twice before the performance. Nevertheless, when Veda reached for the handle, it turned without moving anything else. From my place behind the scenery, I could see the latch sitting stubbornly in its socket.

The visitor knocked again. Veda waited. Her character had spent the previous scene promising to welcome him, and the pause was growing longer than any pause in rehearsal.

“Your welcome is warm,” the visitor said at last.

“Warmer through a closed door,” Veda answered.

A small ripple of laughter moved through the audience. I reached toward the toolbox. Before my fingers found the wrench, Veda dragged a chair to the door and sat facing it. The visitor settled on the step outside. They spoke their next lines through the wood: his requests became more hopeful, her replies more carefully polite. The scene had found purchase again, a foothold from which it could continue.

I left the toolbox closed. Any attempt to force the latch now would have opened more than the door. It would have interrupted the conversation the two actors had made from its refusal to move.

During the interval I freed the latch. Veda tested the handle, and the door swung inward without a sound.

“There,” she whispered. “Now it can be a door again.”

160. What action was planned to occur at the visitor's third knock?

A) The visitor was to leave the stage.

B) Veda was to open the door.

C) The narrator was to close the toolbox.

D) Veda was to move a chair outside.

161. Why does the narrator decide against trying to force the latch during the scene?

A) The actors have ordered the narrator never to repair the door.

B) The visitor has already left, making the door irrelevant.

C) Opening the door would disrupt the new interaction the actors have built around its remaining closed.

D) The audience has asked for all repairs to occur in view.

162. In "The scene had found purchase again," PURCHASE most nearly means:

A) a secure basis for continuing

B) a payment for the door

C) a transfer of ownership

D) a reduction in price

163. What do the actors' lines about a "warm" welcome accomplish?

A) They announce to the audience that the scenery has broken.

B) They explain the technical cause of the stuck latch.

C) They reveal that both actors have forgotten the visitor's identity.

D) They turn the accidental closed door into an apparently intentional expression of the character's guarded welcome.

164. What gives Veda's final statement, "Now it can be a door again," its particular effect?

A) It proves that Veda believed the door had physically changed into another object.

B) It contrasts the door's temporary role in the invented scene with its ordinary practical function after repair.

C) It suggests the repaired door will never be used in another performance.

D) It claims that a working latch would have prevented the entire play from beginning.

Scientific Passage: A Wheel Allowed to Turn One Way

Imagine a wheel whose rim has teeth shaped like uneven hills: each tooth rises gradually on one side and drops steeply on the other. Beside the rim sits a small pivoting arm. A spring presses its tip toward the teeth. Together, these parts can form a ratchet, a mechanism that permits rotation in one direction while resisting the reverse.

Turn the wheel so that a sloping face meets the arm. The slope pushes the arm outward against the spring. Once the tooth passes, the spring draws the tip back toward the rim. The arm drops into the next gap, ready for the following tooth. Repetition produces a series of small clicks rather than one continuous scrape.

Try to turn the wheel backward, however, and a steep face meets the arm. With the arm positioned to block that face rather than ride over it, the contact resists the backward motion. The arm engages the tooth and acts as a stop.

The spring alone does not create the one-way effect. If both faces let the arm rise equally easily, it could pass over teeth in either direction. The unequal faces and the arm's position are essential. Lifting the arm completely clear would also remove the stop, allowing the wheel to turn backward without meeting that barrier.

165. During the permitted rotation, which part of a tooth pushes the arm outward?

A) Its steep blocking face

B) The empty gap behind it

C) The center of the wheel

D) Its gradually sloping face

166. Which statement best summarizes the explanation of the ratchet?

A) Unequal tooth faces work with a positioned, spring-pressed arm to permit one rotation direction and resist the other.

B) A spring makes any toothed wheel turn automatically without an outside force.

C) The clicking sound itself prevents a wheel from moving backward.

D) A wheel can turn only when its teeth are identical on both sides.

167. What would most likely happen if the arm were held completely clear while someone turned the wheel backward?

A) The spring would reverse the shapes of all the teeth.

B) The steep faces would become sloping faces.

C) The arm would no longer provide the described barrier to backward rotation.

D) The wheel would necessarily turn forward without anyone pushing it.

168. In "The arm engages the tooth and acts as a stop," ENGAGES most nearly means:

A) offers a job to

B) promises to marry

C) makes operative contact with

D) holds the interest of

169. Why does the final paragraph consider teeth whose two faces let the arm rise equally easily?

A) The example introduces a history of how the spring was invented.

B) The contrast shows why the spring by itself cannot account for the directional effect.

C) The comparison shows that tooth shape is irrelevant to operation.

D) The example establishes that every toothed wheel must click continuously.

Humanities Passage: Three Dates, Three Claims

Imagine a manuscript inside a cover stamped 1911. A note explains that a binder supplied the cover that year. Farther inside, a copyist has written, "Copied in 1780." The account itself describes a harbor freezing in 1732. Which date belongs on a historical timeline?

The question needs a second question: a timeline of what? The three dates concern three different things. One refers to a binding, another to the making of this particular copy, and the earliest to the event described in its words. Treating them as interchangeable would mix the history of the physical object with the history reported by its text.

The copying note attests to an act of transcription in 1780. It does not establish that the copyist witnessed the freeze, nor does it establish when someone first composed the account. An older text might have been copied many times before reaching these pages. Conversely, a text that names 1732 does not make its surviving paper that old.

Such distinctions help determine what a source can support. The cover may document a later stage of preservation without changing the date claimed for the freeze. Before using any date as evidence, a reader must identify the action to which it is attached: an event, a composition, a copy, or a binding.

170. According to the example, what happened in 1911?

A) A binder supplied the manuscript's cover.

B) The harbor froze.

C) The copyist made the surviving copy.

D) The account was definitely composed for the first time.

171. Which principle does the manuscript example chiefly illustrate?

A) The most visible date always identifies the event described inside.

B) A date's evidentiary meaning depends on which event or stage in a document's history it identifies.

C) Every surviving copy was written by an eyewitness to its subject.

D) A later cover prevents an older text from preserving useful information.

172. Suppose another copy of the same account is reliably dated 1750. Which conclusion would this new evidence support?

A) The 1780 copyist personally witnessed the freeze of 1732.

B) Every detail in the account is necessarily accurate.

C) The account already existed before the surviving copy made in 1780.

D) The cover stamped 1911 must also have been made in 1750.

173. In "The copying note attests to an act of transcription," ATTESTS TO most nearly means:

A) gives instructions for

B) makes an imitation of

C) predicts a future instance of

D) provides evidence of

174. Why does the opening move from the cover's date inward to the copying note and the account's date?

A) It places all the events in chronological order from earliest to latest.

B) It proves that the earliest date must identify the paper's manufacture.

C) It ranks the dates from most reliable to least reliable.

D) It exposes different dated stages within one object before explaining why they must be distinguished.

Section 4: Mathematics

64 questions | Recommended time: 45 minutes

175. Which number can replace u so that n × u = n for EVERY real number n?

A) 0

B) −1

C) 1

D) n

176. What is the value of $\sqrt{((-14)^2)}$?

A) −14

B) 14

C) 196

D) −196

177. A decimal is formed by writing a 1, then one 0, then a 1, then two 0s, then a 1, then three 0s, and so on forever, increasing each zero run by one:
0.101001000100001...
Which statement is correct?

A) It is rational because all its digits follow a rule.

B) It is rational because only two different digits occur.

C) It is irrational because it never terminates or eventually repeats a fixed block.

D) It is an integer because every digit is either 0 or 1.

178. The notation 6! means 1 × 2 × 3 × 4 × 5 × 6. Which listed number is NOT a factor of 6!?

A) 5

B) 7

C) 9

D) 12

179. A function satisfies f(−x) = f(x) for every real x. If f(7) = 12, what is f(−7)?

A) 12

B) −12

C) 7

D) −7

180. A rule sends x to 4x + 11. Which rule, applied to that output, always recovers the original x?

A) Divide by 4, then subtract 11.

B) Add 11, then divide by 4.

C) Subtract 4, then divide by 11.

D) Subtract 11, then divide by 4.

181. For which real values of k does the equation $(k - 2)x = k^2 - 4$ have exactly one real solution for x?

A) Every real k except −2

B) Every real k except 2

C) Every real k except −2 and 2

D) Every real k, including 2

182. What is the least possible value of $x^2 - 6x + 13$ for real x?

A) −5

B) 3

C) 4

D) 13

183. How many points do two distinct parallel planes have in common?

A) 0

B) 1

C) 2

D) Infinitely many

184. A right triangle has perpendicular legs 6 and 3 units long. A square sits at the right-angle vertex with two sides along the legs and its opposite vertex on the hypotenuse. What is the square's side length?

A) 1.5 units

B) 3 units

C) 4 units

D) 2 units

185. Points A and B are at (−2, 0) and (4, 0). Which set consists of ALL points equally far from A and B?

A) the horizontal line y = 0

B) the circle centered at (1, 0) with radius 3

C) the vertical line x = 1

D) the line segment from A to B

186. Temperature readings satisfy C = (5/9)(F − 32). If a temperature falls by 18°F, by how many degrees Celsius does it fall?

A) 10°C

B) 18°C

C) 24°C

D) 50°C

187. During each 20-second cycle, a nozzle sprays for 3 seconds and stays off for 17 seconds. While on, it delivers 2 milliliters per second. How much does it deliver during 2 minutes of complete cycles?

A) 36 milliliters

B) 12 milliliters

C) 18 milliliters

D) 240 milliliters

188. A grower counts the blossoms on EVERY one of the 48 trees in an orchard. The target group is exactly those 48 trees. Which description fits this collection?

A) a sample consisting of only some target trees

B) an experiment that assigns trees to different treatments

C) a prediction made without observing any target trees

D) a census of the target group

189. A fair coin is tossed independently on each trial until the first tail appears or until three heads have appeared, whichever comes first. The complete possible records are T, HT, HHT, and HHH. What are their probabilities, respectively?

A) 1/4, 1/4, 1/4, 1/4

B) 1/2, 1/4, 1/8, 1/8

C) 1/2, 1/4, 1/4, 0

D) 1/8, 1/8, 1/4, 1/2

190. Five real scores have a median of 8. Each score x is replaced by 20 − x. What is the new median?

A) 8

B) 20

C) 28

D) 12

191. For a positive integer n, n! means the product of the integers from 1 through n. What is (8! + 7!)/7!?

A) 1

B) 8

C) 72

D) 9

192. For positive numbers, round an exact halfway value upward. Round 2.46 first to the nearest tenth and then round that result to the nearest whole number. Compare this with rounding 2.46 directly to the nearest whole number. What are the results, respectively?

A) 3 and 2

B) 2 and 2

C) 2 and 3

D) 3 and 3

193. Percentage error is absolute error divided by the true value, times 100%. A length of 2 meters is reported as 2.2 meters; a length of 100 meters is reported as 105 meters. Which statement is correct?

A) The first report has both the larger absolute error and the larger percentage error.

B) The first report has the smaller absolute error but the larger percentage error.

C) The second report has both the smaller absolute error and the smaller percentage error.

D) The two reports have equal percentage errors because both are too large.

194. The variable n counts the number of complete cable reels inspected during a shift. Which value can n have?

A) −2

B) 1/2

C) √2

D) 0

195. A quadratic polynomial has leading coefficient 1 and zeros −2 and 5. Which polynomial is it?

A) $x^2 + 3x - 10$

B) $x^2 - 7x + 10$

C) $x^2 - 3x - 10$

D) $x^2 + 7x + 10$

196. For which real x is $(x + 1)/(x - 2) \geq 1$?

A) $x \geq 2$

B) $x < 2$

C) $x > 2$

D) Every real x except 2

197. A function uses $f(x) = x + 2$ when $x < 0$ and $f(x) = 2x$ when $x \geq 0$. Which inputs give $f(x) = 1$?

A) −1 and 1/2

B) Only −1

C) Only 1/2

D) −1 and 2

198. A triangle has a right angle and two equal legs. Which description must apply?

A) right and scalene

B) acute and isosceles

C) obtuse and equilateral

D) right and isosceles

199. Rectangle A has horizontal coordinates from 0 to 8 and vertical coordinates from 0 to 5. Rectangle B has horizontal coordinates from 3 to 10 and vertical coordinates from 2 to 7. Their sides are parallel to the coordinate axes. What is their overlapping area?

A) 6 square units

B) 25 square units

C) 15 square units

D) 35 square units

200. A triangle has side lengths 9, 10, and 17 units and area 36 square units. A circle inside the triangle touches all three sides. What is its radius?

A) 1 unit

B) 2 units

C) 4 units

D) 18 units

201. A nonstretching thread is fitted along a curved line and then straightened against a ruler. One end is at 4 centimeters and the other at 13 centimeters. How long is the curved line?

A) 4 centimeters

B) 9 centimeters

C) 13 centimeters

D) 17 centimeters

202. A rectangular panel's actual length can be anywhere from 1.9 to 2.1 meters, inclusive, and its actual width anywhere from 2.9 to 3.1 meters, inclusive. No other constraint relates them. What is its greatest possible area?

A) 6.00 square meters

B) 6.21 square meters

C) 6.50 square meters

D) 6.51 square meters

203. A distance counter displays whole meters using three digits. It counts upward from 000 to 999, then resets to 000. It reads 860 at the start and 140 at the end. It resets exactly twice during that interval. How far has it counted, in kilometers?

A) 1.28 kilometers

B) 0.28 kilometers

C) 2.28 kilometers

D) 2.14 kilometers

204. For each of 24 small meteorites, a researcher records both mass and volume. Which display best shows whether these two numerical measurements vary together?

A) a pie chart showing the 24 meteorites as equal portions

B) an alphabetical list containing only the meteorites' names

C) a scatterplot with one point per meteorite, positioned by its mass and volume

D) a bar chart showing only the total number of meteorites

205. In these stem-and-leaf records, the stem is the whole-number part and the leaf is the tenths digit. Sample A has stem 4 with leaves 2, 4 and stem 5 with leaf 1. Sample B has stem 4 with leaves 4, 8 and stem 5 with leaf 0. Which measurement occurs in BOTH samples?

A) 4.2

B) 4.4

C) 4.8

D) 5.1

206. Two fair six-sided dice are rolled independently. What is the probability that the larger of the two numbers is EXACTLY 4, including a tie of 4 and 4?

A) 7/36

B) 1/6

C) 1/4

D) 2/9

207. What is the sum of all the multiples of 6 from 6 through 120, inclusive?

A) 1200

B) 1260

C) 2520

D) 120

208. Which set contains ALL the integers whose reciprocals are also integers?

A) {1}

B) {−1, 1}

C) All nonzero integers

D) All negative integers

209. A number has prime factorization $2^3 \times 5^2$. How many of its positive divisors are perfect squares?

A) 3

B) 6

C) 4

D) 12

210. What is the value of $(1/2 + 1/3) \div (1/2 - 1/3)$?

A) 1

B) 1/5

C) 5/6

D) 5

211. The degree of a polynomial in one variable is its highest exponent with a nonzero coefficient. What is the degree of $6x^{\square} + x^2 - 4$?

A) 3

B) 5

C) 6

D) 2

212. The same real number t satisfies $x = t + 2$ and $y = 2t - 1$. Which equation relates x and y for every possible t?

A) $y = 2x - 5$

B) $y = 2x + 3$

C) $y = x - 3$

D) $y = 2x - 1$

213. Stand P requires 3 metal bars and 1 panel. Stand Q requires 1 metal bar and 2 panels. A workshop has 13 bars and 11 panels. What is the greatest total number of complete stands it can make?

A) 6

B) 8

C) 7

D) 11

214. When $(2x - 3)(x^2 + 5x + 7)$ is expanded, what is the constant term, the term containing no x?

A) −21

B) 21

C) 14

D) −15

215. Which flat shape can be rolled without stretching or twisting, with one pair of opposite edges joined, to form the curved surface of a cylinder open at both ends?

A) A triangle

B) A semicircle

C) A rectangle

D) A regular pentagon

216. A circle has radius 6 units. Two radii form a 90° sector, and a straight chord joins their endpoints on the circle. What is the area between that chord and the smaller arc?

A) 9π + 18 square units

B) 18π − 18 square units

C) 36π − 18 square units

D) 9π − 18 square units

217. What is the shortest distance from the point (8, 0) to the line y = x + 2?

A) 5 units

B) 5√2 units

C) 10 units

D) 10√2 units

218. A road's grade is 100 times its vertical rise divided by its horizontal run, with both lengths in the same unit. A road rises 18 meters over a horizontal run of 0.45 kilometer. What is its grade?

A) 0.04%

B) 40%

C) 4000%

D) 4%

219. A hollow plastic tube is 10 cm long, with outer radius 1 cm and uniform wall thickness 2 mm. It has no end caps. Each cubic centimeter of plastic has a mass of 1.5 grams. What is the tube's mass?

A) 15π grams

B) 9.6π grams

C) 14.4π grams

D) 5.4π grams

220. Which set consists of categories with a natural order from a lower level to a higher level?

A) Service ratings: poor, fair, good, excellent

B) Paint colors: blue, gray, yellow

C) Bolt identification codes: A09, B15, C30

D) Tree species: elm, fir, ash

221. Five distinct pennants labeled A, B, C, D, and E are placed in a uniformly random order. What is the probability that A appears before B and B before C? They need not be adjacent.

A) 1/3

B) 1/10

C) 1/6

D) 1/60

222. A fair spinner has four equal sectors, exactly one red. For 20 independent spins, Jae claims, "There must be exactly five red results." Which response is correct?

A) Five is the expected count, but the observed count can be different.

B) Exactly five is guaranteed because expected counts are always observed.

C) There must be 20 red results because every spin uses a spinner with a red sector.

D) Any red count other than five proves that the spinner is unfair.

223. A student claims that different decimal strings always represent different real numbers. Which pair disproves that claim? Dots indicate an infinite repetition of the shown pattern.

A) 0.121212... and 0.12

B) 1.010101... and 1.01

C) 0.999... and 1

D) 0.5 and 0.05

224. For a real number x, the value 3x + 2 is irrational. Which statement must be true?

A) x is rational.

B) x is an integer.

C) 3x is an integer.

D) x is irrational.

225. How many positive integers n make (n + 12)/n an integer?

A) 4

B) 5

C) 6

D) 12

226. A monomial is a product of a constant and variables raised to nonnegative integer powers. Which expression is a monomial?

A) $7/x$

B) $7x^2y$

C) $x + y$

D) $\sqrt{x}$

227. Two drone-inspection plans charge by flight time m, in minutes. Plan A costs 12 + 3m dollars; Plan B costs 6m dollars. For which flight times is Plan A strictly cheaper?

A) $m < 4$

B) $m > 4$

C) $m \geq 4$

D) $m > 2$

228. A function satisfies f(u + v) = f(u) + f(v) for all real u and v, and f(1) = 6. What is f(5/3)?

A) 6

B) 18

C) 30

D) 10

229. A real number x must satisfy BOTH 2x + 1 > 9 and 3x − 2 ≤ 7. Which statement describes the full solution set?

A) No real value satisfies both inequalities.

B) Every real value greater than 4 satisfies both.

C) Every real value at most 3 satisfies both.

D) Exactly the real values with $3 < x \leq 4$ satisfy both.

230. Which condition must every regular polygon satisfy?

A) All its sides are equal and all its interior angles are equal.

B) Its sides are equal, but its angles may differ.

C) Its angles are equal, but its side lengths may differ.

D) Its opposite sides are parallel, regardless of its other properties.

231. Chords AB and CD intersect at P inside a circle. Their segment lengths satisfy AP × PB = CP × PD. If AP = 3, PB = 8, and CP = 4, what is PD?

A) 2 units

B) 6 units

C) 11 units

D) 12 units

232. A kite has perpendicular diagonals in the ratio 2:3 and an area of 48 square units. Its area equals half the product of its diagonals. How long is the shorter diagonal?

A) 4 units

B) 8 units

C) 12 units

D) 16 units

233. Which 24-hour clock reading represents 12:07 a.m., seven minutes after midnight?

A) 00:07

B) 12:07

C) 24:07

D) 07:00

234. An original map uses 1 centimeter to represent 40 meters. A photocopy enlarges every printed length to 250% of its original length. On the photocopy, 1 centimeter represents how many meters?

A) 16 meters

B) 100 meters

C) 40 meters

D) 10 meters

235. A sonar transducer is 0.6 meter below the water surface. A pulse travels vertically to the bottom and returns after 80 milliseconds. Sound travels 1,500 meters per second in the water. What is the total water depth?

A) 120.6 meters

B) 60 meters

C) 59.4 meters

D) 60.6 meters

236. A modal category has the greatest frequency. Three categories have frequencies X: 4, Y: k, and Z: 6. For which positive integer k are there exactly two modal categories?

A) 4

B) 7

C) 6

D) 9

237. Two archers record hits and attempts at two ranges.
Archer A: near, 9 hits in 10 attempts; far, 30 hits in 60 attempts.
Archer B: near, 72 hits in 90 attempts; far, 4 hits in 10 attempts.
Which comparison of hit percentages is correct?

A) A has the higher percentage at each range and overall.

B) B has the higher percentage at each range and overall.

C) B has the higher percentage at each range; A has the higher percentage overall.

D) A has the higher percentage at each range; B has the higher percentage overall.

238. Three portraits labeled P, Q, and R are each given one of three nameplates marked P, Q, and R. All six assignments are equally likely. What is the probability that no portrait receives its correct nameplate?

A) 1/6

B) 1/2

C) 1/3

D) 5/6

Section 5: Language

60 questions | Recommended time: 30 minutes

239. A style guide requires an apostrophe when forming the plural of a single lowercase letter. Which sentence follows that rule?

A) The word committee contains two t's.

B) The word committee contains two ts.

C) The word committee contains two t's'.

D) The word committee contains two ts'.

240. One speaker's uninterrupted quotation runs for two paragraphs. Under the standard convention for such quotations, where do quotation marks go?

A) Opening and closing marks surround each paragraph separately.

B) One opening mark begins the first paragraph; no mark begins the second; one closing mark ends it.

C) An opening mark begins only the second paragraph; closing marks end both paragraphs.

D) An opening mark begins each paragraph; a closing mark appears only at the end of the second.

241. Which sentence correctly uses a matched pair of dashes to enclose an interruption?

A) The sled—even without a rider, slid down the slope.

B) The sled, even without a rider—slid down the slope.

C) The sled—even without a rider—slid down the slope.

D) The sled—even without a rider slid down the slope.

242. Amira spells her full name Amira de la Vega. Which version preserves that name's capitalization within a sentence?

A) We sent the invitation to Amira De La Vega.

B) We sent the invitation to amira de la Vega.

C) We sent the invitation to Amira de la vega.

D) We sent the invitation to Amira de la Vega.

243. Which version uses the conventional capitalization of a formal letter's general salutation?

A) To whom it may concern:

B) To Whom It May Concern:

C) To Whom it May Concern:

D) to Whom It May Concern:

244. Which spelling correctly completes the sentence?
The pilot began a careful ____ around the storm.

A) manuever

B) maneuvar

C) maneuver

D) manuver

245. Choose the correctly spelled word meaning an object kept as a reminder of a place or experience.

A) souvenier

B) suvenir

C) souvenir

D) souvineer

246. Which word correctly completes the sentence by referring to the entire event in the first clause?
The lock froze, ____ delayed the departure.

A) whom

B) what

C) which

D) that

247. Which sentence correctly expresses two linked comparative changes?

A) The more tightly the cord is twisted, the less easily it bends.

B) The more tightly the cord is twisted, the least easily it bends.

C) The most tightly the cord is twisted, the less easily it bends.

D) The more tightly the cord is twisted, the less easier it bends.

248. The speaker cannot swim now. Which sentence correctly expresses a wish contrary to that present fact?

A) I wish I could swim.

B) I wish I can swim.

C) I wish I will can swim.

D) I wish I am able swim.

249. Which word correctly completes the sentence?
Unexpected questions did not ____ the experienced presenter.

A) phase

B) phrase

C) frame

D) faze

250. Which pair correctly completes the sentence?
The report seemed ____ because its claims had strong evidence; a ____ reader might have believed it without any evidence.

A) credulous; credible

B) credible; credulous

C) credible; credible

D) credulous; credulous

Editing Passage: Three Entries for Three Situations

(1) A cable-testing team uses three different entries in its stretch records. (2) Write 0 only when a completed measurement shows no extension. (3) Write NR when a stretch reading was required but was not recorded. (4) Write NA when that sample was not assigned a stretch test, as with a sample selected only for a color inspection. (5) These entries do not mean the same thing: zero is an observed result, NR marks missing information, and NA marks a test that did not apply. (6) Leaving all three situations blank would hide those differences. (7) A numerical average should treat every NR entry as an observed zero.

251. Which formatting change would make sentences 2–4 easiest to consult as a reference while preserving their meaning?

A) Place the three code entries on separate lines, each followed by its condition.

B) Merge the code entries into one unlabeled string of letters and digits.

C) Remove the conditions and show only the three symbols in a row.

D) List the samples by color and omit the instructions for recording them.

252. Which proposed example would best clarify the distinction between NR and NA after sentence 4?

A) A scheduled test with a lost reading receives NA; every color-inspection sample receives NR.

B) A scheduled test with a lost reading receives NR; a sample assigned only a color inspection receives NA.

C) A completed measurement showing no extension receives NR; a missing required reading receives 0.

D) A missing required reading and an unassigned test both receive 0 because neither provides a number.

253. Which replacement for sentence 7 resolves its conflict with the recording rules and clearly tells a reader how to calculate an average?

A) Average every entry as zero, including all samples for which no stretch test was assigned.

B) Convert every NR entry to NA before averaging, since both are simply missing digits.

C) Discard all observed zeros and average only positive readings, because zero means no measurement occurred.

D) Average only the recorded numerical readings; report required but unrecorded readings separately as NR.

254. Which version correctly places a colon between the speaker label and the spoken words in a script?

A) GUIDE The: tram has stopped.

B) GUID:E The tram has stopped.

C) GUIDE: The tram has stopped.

D) GUIDE The tram has: stopped.

255. Which sentence correctly places a comma to set off the contrasting phrase "not green"?

A) The awning was blue, not green.

B) The awning, was blue not green.

C) The awning was, blue not green.

D) The awning was blue not, green.

256. A technical sheet writes the unit name volt in lowercase and its symbol V in uppercase. Which version follows both conventions?

A) The reading was 6 Volts, written 6 V.

B) The reading was 6 volts, written 6 v.

C) The reading was 6 Volts, written 6 v.

D) The reading was 6 volts, written 6 V.

257. In a fable, Chance is a character's name. Which sentence correctly distinguishes that name from the common noun chance?

A) The crowd listened as Chance asked for a Chance to speak.

B) The crowd listened as Chance asked for a chance to speak.

C) The crowd listened as chance asked for a chance to speak.

D) The crowd listened as chance asked for a Chance to speak.

258. Which spelling correctly completes the sentence?
A covered ____ cooled on the counter.

A) caserole

B) casserolle

C) casserole

D) casserol

259. Choose the correctly spelled word meaning a small-scale version of something.

A) minature

B) miniatuer

C) minniature

D) miniature

260. Which spelling names the behavior of claiming moral standards that one's actions violate?

A) hypocracy

B) hipocrisy

C) hypocricy

D) hypocrisy

261. Which verb form correctly completes this emphatic statement?
I did ____ the latch before leaving.

A) checks

B) check

C) checked

D) checking

262. Which verb correctly treats twelve kilometers as ONE total distance?
Twelve kilometers ____ too far to walk before dawn.

A) are

B) were

C) have been

D) is

263. Which sentence correctly uses a perfect passive infinitive to place the opening before the package's arrival?

A) The package appears to had been opened before it arrived.

B) The package appears to have been opened before it arrived.

C) The package appears to having been opened before it arrived.

D) The package appears to have be opened before it arrived.

264. Which word correctly completes the sentence?
The designer had a ____ for combining unusual colors successfully.

A) flair

B) flare

C) flail

D) flaw

265. Which pair correctly completes the sentence?
The angler ____ the hook while the observer watched with ____ breath.

A) baited; bated

B) bated; baited

C) baited; baited

D) bated; bated

Editing Passage: Following Two Air Routes

(1) A wall vent feeds outside air into an intake duct. (2) The entry tube carries this air to a splitter. (3) The splitter divides the incoming stream into two branches. (4) Dust is caught by a filter as one branch passes through it. (5) This filtered branch then leaves through a low opening near the floor. (6) The other branch passes along an upper channel without going through the filter. (7) Small openings near the ceiling release its air into the room. (8) Describing the two branches separately helps readers follow each stream from the splitter to its own exit.

266. The writer wants sentence 4 to begin with an already established branch before introducing its filter. Which revision best follows that information flow?

A) Dust is caught by a filter as one branch passes through it.

B) A filter catches dust from one of the two branches.

C) One branch passes through a filter that catches dust.

D) The capture of dust occurs when a filter receives one branch.

267. The writer intends entry tube in sentence 2 to name the intake duct already introduced in sentence 1. Which revision best prevents readers from imagining an additional part?

A) Replace The entry tube with The intake duct.

B) Replace The entry tube with A second, separate tube.

C) Replace The entry tube with An unidentified opening.

D) Replace The entry tube with Both intake ducts.

268. Which caption could summarize the described system accurately, without implying that every stream passes through the filter?

A) Both branches pass through the same filter and rejoin before leaving through one low opening.

B) After the splitter, one branch is filtered and exits low; the other bypasses the filter and exits near the ceiling.

C) The splitter sends all air to the ceiling openings first, and the low opening then draws it back into the filter.

D) The wall vent draws air out of the room while the splitter blocks both outgoing branches.

269. Which phrase uses hyphens correctly?

A) a finely-woven belt

B) a finely woven belt

C) a finely woven-belt

D) a finely-woven-belt

270. Which sentence correctly punctuates the academic credential following the name?

A) Marta Dune, MD sent the agenda.

B) Marta Dune MD, sent the agenda.

C) Marta, Dune MD sent the agenda.

D) Marta Dune, MD, sent the agenda.

271. Which phrase correctly uses a possessive construction to express a travel interval of three days?

A) three days' travel

B) three day's travel

C) three days travel

D) three days's travel

272. Which sentence correctly capitalizes the formal name of the time zone?

A) The broadcast begins at noon Eastern Standard Time.

B) The broadcast begins at noon eastern Standard Time.

C) The broadcast begins at noon Eastern standard time.

D) The broadcast begins at noon eastern standard time.

273. Which sentence correctly capitalizes the name of the Dutch city?

A) We changed trains in the Hague.

B) We changed trains in The hague.

C) We changed trains in The Hague.

D) We changed trains in the hague.

274. Which spelling correctly names a small cloth carried for wiping the nose?

A) hankerchief

B) handkerchief

C) handkercheif

D) handkercheef

275. Which spelling correctly names a person employed to drive a passenger car?

A) chauffer

B) chaffeur

C) chauffeur

D) chauffuer

276. Choose the word that correctly completes the sentence in formal written English.
They disagreed about ___ the path was open.

A) if

B) that

C) whether

D) than

277. Choose the verb that correctly completes the sentence.
Each traveler should check whether they ___ carrying the correct pass.

A) is

B) are

C) has

D) be

278. Which sentence indicates that Nola's speaking ended, rather than that she interrupted another activity in order to speak?

A) Nola stopped speaking.

B) Nola stopped to speak.

C) Nola began speaking.

D) Nola continued speaking.

279. Choose the correct word.
The squirrel hid a ___ of acorns inside the hollow trunk.

A) horde

B) herd

C) hoard

D) heard

280. Choose the pair that correctly completes the sentence.
The painter stretched the ___ while the organizers prepared to ___ households about the proposed route.

A) canvass; canvas

B) canvas; canvas

C) canvass; canvass

D) canvas; canvass

281. Choose the pair that correctly completes the sentence.
A guest author contributed the ___; the usher waved the audience ___.

A) foreword; forward

B) forward; foreword

C) foreword; foreword

D) forward; forward

Editing Passage: Six Words, Two Moments

(1) A six-word story should offer an event, not just an inventory of things. (2) A draft reads, "The statue sneezed. The pigeons scattered." (3) Its unexpected action produces a connected response, so the line suggests change. (4) The six words "Tall statues overlook the empty square" mainly describe a settled scene. (5) A tight limit makes each word's job important. (6) One verb can sometimes carry the meaning of a longer phrase. (7) Shortening a line should preserve who acts and what follows from the action. (8) The tiny story need not explain how a stone statue can sneeze in order to make the pigeons' response understandable.

282. Which six-word line best follows the passage's guidance by presenting an event and a connected response?

A) Pigeons rested beside the weathered statue.

B) The square had statues and pigeons.

C) Birds landed; newspapers reached the station.

D) The statue blinked; the pigeons fled.

283. An editor must replace "scattered" in sentence 2 with the two-word phrase "flew away," keep the statue as the one that sneezes, and keep the entire story at six words. Which revision meets all three requirements?

A) The statue sneezed; the pigeons flew.

B) The statue sneezed; pigeons flew away.

C) The statue sneezed; pigeons quickly flew.

D) The pigeons sneezed; statues flew away.

284. A reference to a Bible passage conventionally places a colon between the chapter number and the verse number. Which reference follows that convention?

A) Genesis 2,7

B) Genesis 2;7

C) Genesis 2.7

D) Genesis 2:7

285. Which sentence correctly writes the compound color term?

A) The glaze looked blue, green.

B) The glaze looked bluegreen.

C) The glaze looked blue-green.

D) The glaze looked blue; green.

286. Use sentence-style capitalization for this title: capitalize its first word and the person's name, rather than every principal word. Which version follows that style?

A) Crossing the desert with Elena

B) Crossing the Desert with Elena

C) crossing the desert with Elena

D) Crossing the desert with elena

287. Which sentence correctly capitalizes the hyphenated historical adjective?

A) The carvings are pre-Columbian.

B) The carvings are Pre-Columbian.

C) The carvings are pre-columbian.

D) The carvings are Pre-columbian.

288. Which spelling correctly names an elected county law-enforcement officer?

A) sherriff

B) sheriff

C) sherif

D) sherrif

289. Which spelling correctly names a device or technique that helps a person remember information?

A) mnemonic

B) mneumonic

C) mnemonick

D) nemmonic

290. Which spelling correctly names a person with expert appreciation of a particular art or field?

A) connoiseur

B) connoisseur

C) connoissour

D) conosseur

291. Choose the pair of verbs that correctly completes the sentence.
A number of lenses ___ to be missing, but the number of missing lenses ___ unchanged.

A) seem; remain

B) seems; remains

C) seems; remain

D) seem; remains

292. Choose the phrase that correctly expresses "if the weather allows it" at the beginning of this sentence.
___, the balloon flight will begin at dawn.

A) Weather permits

B) Weather to permit

C) Weather has permitted

D) Weather permitting

293. Choose the verb form required by the phrase "had better."
You had better ___ for the earlier ferry.

A) to leave

B) leaving

C) leave

D) left

294. Choose the correct word.
Before choosing a route, the navigator would ___ over the chart for several minutes.

A) pour

B) poor

C) pore

D) paw

295. Choose the pair that correctly completes the sentence.
The chemist sealed the sample in a ___ and described its smell as ___.

A) vile; vial

B) vial; vial

C) vial; vile

D) vile; vile

296. Choose the pair that correctly completes the sentence.
One inspector was so ___ about his own method that he ignored warnings; the other was so ___ that she agreed to every request.

A) complaisant; complacent

B) complacent; complaisant

C) complacent; complacent

D) complaisant; complaisant

Editing Passage: A Definition Needs an Edge

(1) A definition for beginners should name a general class and then supply the feature that distinguishes the item within that class. (2) For example, a cantilever is a beam fixed at one end and unsupported at the other. (3) Beam identifies the kind of object, while the support arrangement sets it apart. (4) A bridge may contain cantilevers, but a cantilever need not be an entire bridge. (5) A projecting balcony can also use this arrangement. (6) Definitions can therefore be more useful than a list of familiar examples. (7) An example helps readers picture something, but the distinguishing feature helps them recognize another instance they have never seen. (8) A useful definition marks a boundary as well as naming a group.

297. Use the passage's definition method with these facts: a wicket gate is a small gate set within a larger gate, allowing passage without opening the larger one. Which sentence best uses both a general class and a distinguishing feature?

A) A wicket gate is something people may notice at an entrance.

B) A wicket gate is a gate known as a wicket gate.

C) A wicket gate is not a cantilever or a projecting balcony.

D) A wicket gate is a small gate within a larger gate that permits passage without opening the larger one.

298. Which sentence added after sentence 2 would most clearly establish the definition's boundary by providing a contrasting nonexample?

A) A beam resting on supports at both ends is not a cantilever, even when its middle has no support.

B) A cantilever must always be a complete bridge.

C) Any beam with an unsupported middle is therefore a cantilever.

D) Every type of beam can be defined only by listing places where it appears.

Answer Key

Question numbers continue across all five sections.

Verbal Skills

1. D	2. B	3. A	4. B	5. A	6. C	7. C	8. C
9. B	10. D	11. A	12. D	13. A	14. B	15. C	16. C
17. D	18. B	19. D	20. A	21. A	22. B	23. B	24. D
25. B	26. D	27. C	28. A	29. C	30. C	31. D	32. A
33. A	34. A	35. D	36. C	37. C	38. B	39. D	40. B
41. D	42. C	43. A	44. C	45. B	46. C	47. C	48. A
49. B	50. B	51. D	52. B	53. C	54. D	55. C	56. B
57. D	58. A	59. A	60. D				

Quantitative Skills

61. B	62. A	63. D	64. B	65. C	66. A	67. B	68. C
69. D	70. B	71. A	72. C	73. D	74. C	75. A	76. B
77. D	78. C	79. A	80. B	81. C	82. D	83. B	84. A
85. C	86. D	87. A	88. D	89. B	90. C	91. D	92. A
93. B	94. B	95. C	96. A	97. D	98. D	99. C	100. D
101. C	102. A	103. B	104. C	105. D	106. A	107. A	108. B
109. D	110. A	111. B	112. C				

Reading Comprehension

113. A	114. D	115. D	116. A	117. B	118. A	119. D	120. B
121. C	122. B	123. D	124. C	125. C	126. A	127. B	128. C
129. D	130. C	131. D	132. B	133. C	134. B	135. A	136. D
137. A	138. A	139. B	140. C	141. A	142. C	143. D	144. B
145. D	146. A	147. A	148. D	149. C	150. B	151. B	152. B
153. C	154. A	155. A	156. D	157. C	158. B	159. D	160. B
161. C	162. A	163. D	164. B	165. D	166. A	167. C	168. C
169. B	170. A	171. B	172. C	173. D	174. D		

Mathematics

175. C	176. B	177. C	178. B	179. A	180. D	181. B	182. C
183. A	184. D	185. C	186. A	187. A	188. D	189. B	190. D
191. D	192. A	193. B	194. D	195. C	196. C	197. A	198. D
199. C	200. B	201. B	202. D	203. A	204. C	205. B	206. A

207. B	208. B	209. C	210. D	211. B	212. A	213. C	214. A
215. C	216. D	217. B	218. D	219. D	220. A	221. C	222. A
223. C	224. D	225. C	226. B	227. B	228. D	229. A	230. A
231. B	232. B	233. A	234. A	235. D	236. C	237. D	238. C

Language

239. A	240. D	241. C	242. D	243. B	244. C	245. C	246. C
247. A	248. A	249. D	250. B	251. A	252. B	253. D	254. C
255. A	256. D	257. B	258. C	259. D	260. D	261. B	262. D
263. B	264. A	265. A	266. C	267. A	268. B	269. B	270. D
271. A	272. A	273. C	274. B	275. C	276. C	277. B	278. A
279. C	280. D	281. A	282. D	283. B	284. D	285. C	286. A
287. A	288. B	289. A	290. B	291. D	292. D	293. C	294. C
295. C	296. B	297. D	298. A				

Detailed Explanations

Verbal Skills

1. HSPT-V14-001 - Answer D: an entrance hall
A foyer is an entrance hall or lobby through which people pass when entering a building, rather than an outdoor or storage area.

2. HSPT-V14-002 - Answer B: a small amount of knowledge
A smattering is a small or slight amount of knowledge, insufficient to suggest thorough understanding but greater than complete ignorance.

3. HSPT-V14-003 - Answer A: tension that is readily noticeable
Palpable can describe something so evident that it seems almost touchable; here it makes the tension readily noticeable rather than literally solid.

4. HSPT-V14-004 - Answer B: increasing
Waning interest is decreasing in strength or amount; increasing interest moves in the opposite direction, while the other choices continue the decline.

5. HSPT-V14-005 - Answer A: reactive
In the stated chemical sense, inert means not readily reacting; reactive identifies the opposite tendency rather than a different physical appearance.

6. HSPT-V14-006 - Answer C: carefully considered
An offhand judgment is casual and unconsidered; carefully considered directly reverses that lack of reflection without merely changing how the judgment is delivered.

7. HSPT-V14-007 - Answer C: passage
A ransom is demanded as the price of release; a toll is charged as the price of passage, pairing each payment with what it obtains.

8. HSPT-V14-008 - Answer C: beginning
Intercepting prevents a message from reaching its destination; preempting prevents an anticipated action from beginning, identifying the stage each intervention forestalls.

9. HSPT-V14-009 - Answer B: rule
A concession reduces what is demanded; an exception sets aside a rule for a particular case, reducing the force of an otherwise applicable requirement.

10. HSPT-V14-010 - Answer D: finale
A finale is a concluding part; a hiatus, lull, or respite is a temporary break during which activity or difficulty subsides.

11. HSPT-V14-011 - Answer A: lending; borrowing
Lending and borrowing describe opposite sides of a temporary transfer with an obligation to return the item, unlike the other action pairs.

12. HSPT-V14-012 - Answer D: wind: moving air; to turn a handle
Wind has a short i for moving air and a long i for turning; each other listed pair keeps the same pronunciation across meanings.

13. HSPT-V14-013 - Answer A: The reasoning changes the meaning of the same word between statements.
Fine names a monetary penalty in the sign but means excellent in the response; changing meanings breaks the supposed logical connection.

14. HSPT-V14-014 - Answer B: The absence of disproof alone does not establish that the claim is true.
Failing to disprove a claim leaves its truth unsettled unless further evidence supports it; absence of refutation alone does not prove the claim.

15. HSPT-V14-015 - Answer C: The original cell is faulty.
With a working bulb installed, darkness identifies the cell as faulty; the exactly-one-fault condition then rules out a faulty original bulb.

16. HSPT-V14-016 - Answer C: showing initiative and determination
Gumption combines the initiative to act with the resolve to keep going; it does not mean avoiding difficulty or waiting passively.

17. HSPT-V14-017 - Answer D: a minor personal weakness
A foible is a small weakness or failing in a person's character, not necessarily a serious vice or a deliberate wrongdoing.

18. HSPT-V14-018 - Answer B: the points that stand out as most important
Salient points are prominent or especially important; the term concerns their significance rather than their position, numerical form, or omission.

19. HSPT-V14-019 - Answer D: gloomy
Jovial describes a cheerful, good-humored mood; gloomy reverses that positive mood, while the other choices express similar cheerfulness.

20. HSPT-V14-020 - Answer A: resistant to damage
Susceptible means readily affected or harmed; resistant describes the opposite tendency to withstand the same damaging influence.

21. HSPT-V14-021 - Answer A: made jointly by the parties
Unilateral describes one party acting alone; a decision made jointly shares the deciding role rather than merely changing when or how it is announced.

22. HSPT-V14-022 - Answer B: work
A boycott applies pressure by withholding purchases; a strike applies pressure by withholding labor, pairing each coordinated action with the cooperation withdrawn.

23. HSPT-V14-023 - Answer B: location
A glossary supplies meanings, while an index supplies locations such as page numbers; each reference feature answers a different kind of lookup question.

24. HSPT-V14-024 - Answer D: verb
An adjective can describe a noun, while an adverb can describe how a verb's action occurs; the relationship concerns modification of a grammatical head.

25. HSPT-V14-025 - Answer B: mishap
A mishap is an unfortunate accident, whereas a strategy, policy, or scheme is a plan or guiding course of action.

26. HSPT-V14-026 - Answer D: a soft brush; brush away the dust
Brush names an implement in the first phrase and an action in the second; each alternative retains the same word class in both phrases.

27. HSPT-V14-027 - Answer C: priceless; invaluable
Priceless and invaluable describe extraordinary value despite their negative-looking forms; the alternatives express ordinary negation, absence, or unfavorable judgment.

28. HSPT-V14-028 - Answer A: An additional condition was introduced after the announced condition was met.
The announced sufficient condition was readable labels; refusing approval solely for a newly introduced border requirement changes the stated standard after it is satisfied.

29. HSPT-V14-029 - Answer C: It replaces a limited proposal with a much broader position that was not proposed.
The original permits conversation during a specific pause; the response changes both permission to a demand and the permitted time to the entire performance.

30. HSPT-V14-030 - Answer C: Do not write anything.
Rae's assignment is to reproduce the card's wording; the instruction mentioned inside that wording does not replace the outer copying instruction.

31. HSPT-V14-031 - Answer D: a strong liking
A penchant is a strong preference or inclination, so this person particularly enjoys intricate puzzles.

32. HSPT-V14-032 - Answer A: an ethical reservation
A scruple is a hesitation based on moral concerns; Lena questions whether accepting the credit would be right.

33. HSPT-V14-033 - Answer A: a confident, stylish manner
Panache describes a spirited, stylish display of confidence, rather than indifference, reluctance, or mere technical accuracy.

34. HSPT-V14-034 - Answer A: the highest point
Nadir names the lowest point, so its opposite is the highest point on the same scale.

35. HSPT-V14-035 - Answer D: unrelated to the topic
Germane means relevant or closely related to a subject; an unrelated remark has the opposite relationship.

36. HSPT-V14-036 - Answer C: willing to reconsider firmly held beliefs
A dogmatic person treats beliefs as unquestionable; willingness to reconsider them expresses the opposite intellectual attitude.

37. HSPT-V14-037 - Answer C: location
Forced perspective alters apparent size; ventriloquism alters the apparent location of a voice while its actual source remains elsewhere.

38. HSPT-V14-038 - Answer B: color
A sieve sorts particles by size; a prism spreads white light into its component colors.

39. HSPT-V14-039 - Answer D: height
Doorway width restricts horizontal clearance, while ceiling height restricts vertical clearance for an object passing underneath.

40. HSPT-V14-040 - Answer B: entry
Entry names movement across a boundary into a place; the other words describe remaining or being there.

41. HSPT-V14-041 - Answer D: scissors; tweezers
Scissors and tweezers each name one instrument using a plural form; the other pairs include ordinary countable objects.

42. HSPT-V14-042 - Answer C: Reverse the entire row; exchange its first and last letters.
Reversing twice and exchanging the endpoints twice both undo themselves. Sorting, rotating once per application, and duplicating do not.

43. HSPT-V14-043 - Answer A: The length of time a practice has existed does not establish that it is the best possible practice.
Long use establishes a tradition, but it does not compare that practice with possible improvements or establish optimality.

44. HSPT-V14-044 - Answer C: At least one crew member can both row and sail.
Two disjoint groups cannot each contain more than half the crew; therefore the groups must share at least one member.

45. HSPT-V14-045 - Answer B: Every card accepted under the former rule is accepted under the revised rule.
A blue square satisfies both conditions and therefore also satisfies the revised rule requiring at least one of them.

46. HSPT-V14-046 - Answer C: takes unnecessary time moving along
To dawdle is to move or spend time unnecessarily slowly, delaying progress toward the exit.

47. HSPT-V14-047 - Answer C: an embarrassing social mistake
A gaffe is an awkward social blunder, such as mistakenly addressing a guest by the wrong name.

48. HSPT-V14-048 - Answer A: favoring family members in appointments
Nepotism is favoritism toward relatives, especially when jobs or other advantages are awarded because of family connections.

49. HSPT-V14-049 - Answer B: strict enforcement
Lax enforcement is loose and insufficiently strict; strict enforcement requires close attention to whether rules are followed.

50. HSPT-V14-050 - Answer B: tastefully elegant
Tawdry combines showiness with cheap appearance; tasteful elegance contrasts with that lack of refinement rather than merely reducing brightness.

51. HSPT-V14-051 - Answer D: unable to pay debts
A financially solvent person or business can meet debts; inability to pay them expresses the opposite financial condition.

52. HSPT-V14-052 - Answer B: pressure
Both lines connect points sharing one measured value: contour lines use elevation, while isobars use air pressure.

53. HSPT-V14-053 - Answer C: a sounded note
Both preserve an interval while leaving out the expected content: a printed letter in the text or a sounded note in the music.

54. HSPT-V14-054 - Answer D: water volume passing per unit of time
The relationship contrasts an accumulated total with its rate of change; flow rate measures volume passing per unit of time.

55. HSPT-V14-055 - Answer C: at most one item
At most one permits either zero or one. Each other expression requires a positive number of items.

56. HSPT-V14-056 - Answer B: a rescue boat
A rescue boat helps people after danger arises; the other items are barriers intended to interrupt a harmful path beforehand.

57. HSPT-V14-057 - Answer D: custom/customs; manner/manners
Customs can mean a border-control agency, and manners can mean polite conduct; these are specialized meanings beyond several customs or manners.

58. HSPT-V14-058 - Answer A: Evidence connecting the forecast to the conditions it predicts.
Counting believers measures popularity, not predictive accuracy; evidence about the predicted conditions is needed to support the conclusion.

59. HSPT-V14-059 - Answer A: Records of wooden vessels that sank and therefore could not enter the study.
Selecting only surviving vessels excludes the failures needed to estimate sinking frequency, so survival in the sample cannot establish general reliability.

60. HSPT-V14-060 - Answer D: That a property of each separate component must also hold for the assembled whole.
Assembling strands changes how a load is shared; the ease of breaking one strand does not establish the strength of the complete rope.

Quantitative Skills

61. HSPT-Q14-001 - Answer B: 814
Moving the first digit of 481 to the end leaves 81 followed by 4, giving 814 and restarting the same rotation.

62. HSPT-Q14-002 - Answer A: 5
Since 13 = 2 × 6 + 1, use a(6) + a(7). The displayed sixth and seventh terms are 2 and 3, giving 5.

63. HSPT-Q14-003 - Answer D: 14
Group by the number of pairs inside the first matched pair: the four cases contribute 1 × 5, 1 × 2, 2 × 1, and 5 × 1, totaling 14.

64. HSPT-Q14-004 - Answer B: 10
After 8, the unused numbers 1, 5, and 7 share no factor greater than 1 with 8; the least eligible unused integer is 10.

65. HSPT-Q14-005 - Answer C: 8
The alternating digit sum is 6 − d + 2 = 8 − d. For a digit d, its only possible multiple of 11 is 0, so d = 8.

66. HSPT-Q14-006 - Answer A: 1,001
If the original integer is n, repeating its three digits produces 1,000n + n = 1,001n; division by n therefore gives 1,001.

67. HSPT-Q14-007 - Answer B: 2
The first three expressions all equal 0.125; the last two both equal 0.0125, leaving exactly two distinct values after equivalent representations are grouped.

68. HSPT-Q14-008 - Answer C: The two quantities are equal.
Exchanging the names A and B pairs every arrangement of AAB with exactly one arrangement of ABB; each set has three members.

69. HSPT-Q14-009 - Answer D: The relationship cannot be determined from the information given.
For n = 6, the quantities are 2 and 1; for n = 8, they are 1 and 3, so their ordering can reverse.

70. HSPT-Q14-010 - Answer B: Quantity B is greater.
The real solutions are −3 and 3. Only 3 is positive, giving one solution for Quantity A and two for Quantity B.

71. HSPT-Q14-011 - Answer A: Quantity A is greater.
An exterior angle equals the sum of the two nonadjacent interior angles; both are positive, so their sum exceeds either one.

72. HSPT-Q14-012 - Answer C: The two quantities are equal.
Every prism has volume equal to base area times perpendicular height; congruent bases and equal perpendicular heights make the two volumes equal.

73. HSPT-Q14-013 - Answer D: The relationship cannot be determined from the information given.
The common chord may pass through the smaller circle's center and equal its diameter, or miss that center and be shorter; the information does not specify which.

74. HSPT-Q14-014 - Answer C: 17
The next integer, 16, is divisible by the square 4; 17 has no square factor greater than 1 and is therefore the next entry.

75. HSPT-Q14-015 - Answer A: 1,807
The product of the first three terms is 42, so the next calculation is 42 × 43 + 1 = 1,807.

76. HSPT-Q14-016 - Answer B: 46
The merges leave 6, 7, 13, 20; then 13, 13, 20; then 20, 26. The next two smallest numbers sum to 46.

77. HSPT-Q14-017 - Answer D: 9:10
Matching the b terms gives a:b = 9:12 and b:c = 12:10, so the shared middle quantity yields a:c = 9:10.

78. HSPT-Q14-018 - Answer C: 12
For b = 2 or 3, b^3 is not a square. At b = 4, a = 8 and the sum is 12; larger b gives a larger sum.

79. HSPT-Q14-019 - Answer A: 14
Write the first quotient as 3k + 2. The original integer is 5(3k + 2) + 4 = 15k + 14, leaving remainder 14.

80. HSPT-Q14-020 - Answer B: 10
The number 10 equals $1^2 + 3^2$. The other listed numbers require equal squares, a zero square, or do not have a qualifying representation.

81. HSPT-Q14-021 - Answer C: The two quantities are equal.
The original fourth ordered value becomes positions 10 through 12 in the 21-entry list; its middle, position 11, therefore has the same value.

82. HSPT-Q14-022 - Answer D: The relationship cannot be determined from the information given.
For a = b = 1.4, the quantities are 2 and 3; for a = b = 1.6, they are 4 and 3, so either can be greater.

83. HSPT-Q14-023 - Answer B: Quantity B is greater.
Removing blue leaves the positive red count unchanged but decreases the total by one, so the red fraction has a smaller denominator and becomes larger.

84. HSPT-Q14-024 - Answer A: Quantity A is greater.
Both rectangles have the cube's height, but the slice's width is a square-base diagonal, longer than the face's width of one cube edge.

85. HSPT-Q14-025 - Answer C: The two quantities are equal.
Triangles ABM and ACM have three matching side lengths: AB = AC, BM = CM, and shared AM. Their matching apex angles are therefore equal.

86. HSPT-Q14-026 - Answer D: The relationship cannot be determined from the information given.
Equal areas give equal heights h. If the altitude feet coincide, CD = 2h; if they differ, CD is a longer diagonal, so equality is not guaranteed.

87. HSPT-Q14-027 - Answer A: 8
The rule gives 2 × 39 + 2 = 80 and 2 × 8 + 0 = 16; next, 2 × 1 + 6 = 8.

88. HSPT-Q14-028 - Answer D: 1863
After 1836, the remaining arrangement beginning with 18 is 1863; numbers beginning with 3 come later.

89. HSPT-Q14-029 - Answer B: 1100
The next move is L. In 0100, the rightmost 1 is the second digit, so flip the first digit to obtain 1100.

90. HSPT-Q14-030 - Answer C: 32
A capacity of 16 cannot hold 17 items. The next available capacity is 32, so the seventeenth term is 32.

91. HSPT-Q14-031 - Answer D: 500
Divide 400,000 by 800 to obtain 500; equivalently, 4 ÷ 8 times 10^3 equals 500.

92. HSPT-Q14-032 - Answer A: 32
Let the original counts be 3k and 5k. Then 3k + 4 = 5k − 4, giving k = 4 and 8k = 32 threads.

93. HSPT-Q14-033 - Answer B: $\sqrt{5} + 2$
Multiply numerator and denominator by $\sqrt{5} + 2$. The denominator becomes 5 − 4 = 1, leaving $\sqrt{5} + 2$.

94. HSPT-Q14-034 - Answer B: Quantity B is greater.
Complement probabilities are 1 − P(E) and 1 − P(F). Subtracting the larger original probability from 1 gives the smaller complement.

95. HSPT-Q14-035 - Answer C: The two quantities are equal.
Multiplication by 4 multiplies both the range and mean by 4. Their ratio stays unchanged, and the positive mean permits division.

96. HSPT-Q14-036 - Answer A: Quantity A is greater.
The expression in Quantity B is a weighted average with positive weights. It lies strictly between the distinct numbers, below the larger one.

97. HSPT-Q14-037 - Answer D: The relationship cannot be determined from the information given.
The folded rectangle can measure 7 by 6, with perimeter 26, or 3.5 by 12, with perimeter 31. These fall on opposite sides of 28.

98. HSPT-Q14-038 - Answer D: The relationship cannot be determined from the information given.
The triangle 10–24–26 has longer leg 24, while 15–20–25 has longer leg 20. Both have perimeter 60, but the comparisons differ.

99. HSPT-Q14-039 - Answer C: The two quantities are equal.
Vertically opposite angles are formed by the same two intersecting lines and always have equal measures.

100. HSPT-Q14-040 - Answer D: 180
After subtracting 1/2 and 1/3, the remainder is 7/60. Subtracting 1/9 leaves 1/180, so the next denominator is 180.

101. HSPT-Q14-041 - Answer C: 123456
Split 246135 into 246 and 135. Taking alternating digits from 135 and 246 gives 1, 2, 3, 4, 5, 6.

102. HSPT-Q14-042 - Answer A: 123112123
Replace the six digits of 123112 by 12, 3, 1, 12, 12, and 3, then join those blocks to obtain 123112123.

103. HSPT-Q14-043 - Answer B: −6
Multiplying −6 by itself three times gives −216; an odd power preserves the negative sign.

104. HSPT-Q14-044 - Answer C: 999
Each aligned pair of digits sums to 9. Thus the hundreds, tens, and units contribute 900, 90, and 9, totaling 999.

105. HSPT-Q14-045 - Answer D: 7
The numeral represents two groups of b and one unit, so 2b + 1 = 15. Therefore b = 7.

106. HSPT-Q14-046 - Answer A: 12
No integer p lies strictly between 2q/5 and 3q/7 for q from 1 through 11. For q = 12, p = 5 works.

107. HSPT-Q14-047 - Answer A: Quantity A is greater.
Both fractions have positive numerator W. Since L is smaller than W + L, division by L produces the greater value.

108. HSPT-Q14-048 - Answer B: Quantity B is greater.
Multiplication by x, a positive factor smaller than 1, decreases x. Consequently x^2 is smaller than x.

109. HSPT-Q14-049 - Answer D: The relationship cannot be determined from the information given.
For n = 2, the quantities are 2 and 1. For n = 6, they are 720 and 729. The ordering therefore changes.

110. HSPT-Q14-050 - Answer A: Quantity A is greater.
In one circle, a chord nearer the center is longer. The 2-unit distance leaves a longer half-chord than the 5-unit distance.

111. HSPT-Q14-051 - Answer B: Quantity B is greater.
Opposite interior angles of a quadrilateral inscribed in a circle sum to 180°, which is less than 200°.

112. HSPT-Q14-052 - Answer C: The two quantities are equal.
Parallel straight lines maintain a constant perpendicular distance, so changing the measurement location does not change their separation.

Reading Comprehension

113. HSPT-R14-001 - Answer A: the words "the only things"
Lina keeps the questions but draws a line through the phrase that treats them as the exclusive measures of what she should know.

114. HSPT-R14-002 - Answer D: accused
The letter makes Lina feel judged for falling behind its expectations, so charged means accused rather than paid or electrically supplied.

115. HSPT-R14-003 - Answer D: They show that meaningful learning occurred outside her earlier predictions.
Those unlisted accomplishments challenge the word only: Lina values what she learned without claiming that the original questions no longer matter.

116. HSPT-R14-004 - Answer A: It contrasts her younger certainty with her older, more questioning response.
The handwriting embodies the younger Lina's certainty, which the older Lina later questions when she reconsiders the letter's narrow expectations.

117. HSPT-R14-005 - Answer B: someone whose growth cannot be fully prescribed by her present expectations
The imagined future person continues the letter's conversation across ages, while the blank paper leaves that person space for developments Lina cannot yet predict.

118. HSPT-R14-006 - Answer A: The buried shell dissolves while the surrounding rock remains.
Water dissolves the shell after the surrounding sediment has hardened, leaving an empty impression in rock that remains in place.

119. HSPT-R14-007 - Answer D: New sediment or minerals fill the mold and harden.
The third paragraph explains that filling the mold with sediment or minerals, followed by hardening, can produce a solid cast of the shell.

120. HSPT-R14-008 - Answer B: Molds and casts can preserve aspects of a shell without preserving the original shell itself.
The passage follows the loss of a shell and distinguishes hollow molds from solid casts, both capable of preserving shape without original shell material.

121. HSPT-R14-009 - Answer C: The ridge pattern may be studied even though the shell itself is gone.
An external mold preserves outer-surface impressions before any filling occurs; retained ridges provide shape evidence even after the original shell has dissolved.

122. HSPT-R14-010 - Answer B: a hollow space
The shell dissolves and leaves an inward empty form surrounded by rock, making cavity a hollow space rather than the later solid cast.

123. HSPT-R14-011 - Answer D: a fortunate draw
The opening paragraph states that a fortunate draw removes the danger after the second player has already made a careless move.

124. HSPT-R14-012 - Answer C: A choice should be judged using what was reasonably knowable when it was made, as well as a careful view of later results.
The passage separates decision quality from a single outcome while retaining visible risks and longer outcome records as relevant evidence.

125. HSPT-R14-013 - Answer C: The player ignored a danger clearly visible on the board before choosing.
An avoidable risk visible at the time counts against a decision; the passage does not require knowledge of concealed future events.

126. HSPT-R14-014 - Answer A: justify
Warrant means provide sufficient justification: one lucky draw does not supply a sound basis for calling an otherwise careless decision wise.

127. HSPT-R14-015 - Answer B: To prevent the main distinction from being stretched into excuses for every mistake or rejection of all results
The third paragraph limits the argument: visible avoidable risks remain relevant, and repeated outcomes can reveal weaknesses that one result cannot establish.

128. HSPT-R14-016 - Answer C: It separates information available at the choice from information revealed afterward.
The screen is a conceptual boundary between choosing and learning the result, helping a reviewer avoid treating later information as previously available.

129. HSPT-R14-017 - Answer D: SARA
The passage explicitly shows SARA on the first tag before the clerk tears it off and writes Soraya's full name on another.

130. HSPT-R14-018 - Answer C: She discovers that asking for her name to be correct can be a small, reasonable request.
Soraya expects correction to cause a major disruption, but the brief, ordinary exchange gives her room to ask for a name that actually belongs to her.

131. HSPT-R14-019 - Answer D: She had anticipated impatience that the scene does not show.
Soraya imagines a sigh, but the waiting person is simply reading a notice, providing no evidence for the impatient reaction she feared.

132. HSPT-R14-020 - Answer B: It presents accepting approximate names as an established habit across several situations.
The accumulated settings extend the behavior beyond this single visit, showing a recurring habit without attributing deliberate hostility to the people involved.

133. HSPT-R14-021 - Answer C: It transforms the idea of holding from causing delay into making room for someone.
The repeated word shifts from obstructing others to accommodating Soraya, expressing how the small correction feels more welcoming than disruptive.

134. HSPT-R14-022 - Answer B: a lower part
The second paragraph traces a top-of-object ray straight through the opening to a lower region of the opposite inside surface.

135. HSPT-R14-023 - Answer A: on the inside surface opposite the hole
The opening sentence locates the image on the inside surface across from the hole, and the remaining explanation traces light toward that surface.

136. HSPT-R14-024 - Answer D: Restricting straight light paths helps form a separated, inverted image.
Straight paths explain inversion, while restricting their spread limits overlapping patches; together these ideas explain image formation rather than brightness alone.

137. HSPT-R14-025 - Answer A: allowed to enter
The surrounding paragraph describes light entering through an opening, so admitted means allowed inside rather than confessed or established as true.

138. HSPT-R14-026 - Answer A: Each opening can project its own point mapping, so the resulting images may overlap.
Each opening admits its own straight paths to the opposite surface; the passage's explanation of overlapping contributions supports potentially overlapping projected images.

139. HSPT-R14-027 - Answer B: wearing away by flowing water
The first paragraph contrasts water wearing away an outer bank with sediment accumulating along the inner bank of the bend.

140. HSPT-R14-028 - Answer C: A changing river can abandon a loop that preserves evidence of its former course.
The passage explains a shortcut, separation of the old loop, and the remaining shape as evidence of a course the active river no longer follows.

141. HSPT-R14-029 - Answer A: It may occupy an abandoned loop that later partly filled in.
The passage describes an abandoned curved lake gradually filling with sediment and plant material, making a horseshoe-shaped marsh a possible later stage.

142. HSPT-R14-030 - Answer C: the narrow strip of land between neighboring arms of the loop
The passage directly applies neck to the narrow land separating nearby arms of one loop, not to the water body or the whole channel.

143. HSPT-R14-031 - Answer D: It traces how a shortcut can redirect flow and leave the old bend separated.
The paragraph links a break through the neck to redirected flow, reduced flow in the old loop, and sediment blocking the loop's entrances.

144. HSPT-R14-032 - Answer B: A bend may narrow, flow may cut across its neck, and the abandoned loop may later become a lake that gradually fills.
The correct summary keeps the narrowing, cutoff, separation, and possible filling in order while preserving the passage's qualified language about these changes.

145. HSPT-R14-033 - Answer D: Alphabetically by destination
The opening paragraph explicitly says the clerks first arranged the parcels alphabetically by destination before Kelm requested other arrangements.

146. HSPT-R14-034 - Answer A: Perfecting an arrangement can become a substitute for accomplishing the work the arrangement should serve.
Kelm repeatedly perfects the room while preventing any delivery, confusing preparation for the actual purpose of shipping.

147. HSPT-R14-035 - Answer A: Letting a driver remove one parcel from an otherwise complete row.
Kelm objects that removing a parcel would leave a gap, so dispatching even one parcel directly violates his stated priority.

148. HSPT-R14-036 - Answer D: It uses formal language of progress for a change that has made destinations harder to find.
The dignified phrase treats the less useful arrangement as an achievement, producing humor through understated praise of an absurd result.

149. HSPT-R14-037 - Answer C: It allows readers to recognize a failure that Kelm presents as a success.
Readers see that nothing has been delivered, while Kelm celebrates complete readiness; the contrast creates irony about his definition of success.

150. HSPT-R14-038 - Answer B: Peeling an orange beside an open window
The first sentence places the narrator beside the open window peeling an orange when the first white object strikes.

151. HSPT-R14-039 - Answer B: A brief hailstorm overwhelms an ordinary moment before the narrator resumes a simple activity.
The passage moves from peeling an orange through the intense hailstorm and back to the same ordinary activity.

152. HSPT-R14-040 - Answer B: The storm continues in another place after its strongest effects have passed the narrator's house.
The different roof sounds show that the intense hail has moved beyond the narrator's house while continuing down the road.

153. HSPT-R14-041 - Answer C: used up
The hard blows are ending and soft dripping follows, so spent means that the storm's local force is used up.

154. HSPT-R14-042 - Answer A: It frames the intense storm as an interruption within an ordinary activity that continues.
Repeating the unfinished activity connects the ending to the opening and emphasizes the storm's brief interruption of ordinary life.

155. HSPT-R14-043 - Answer A: Radiation
The third paragraph explicitly states that radiation crosses empty space without moving air or a continuous material connection.

156. HSPT-R14-044 - Answer D: Several design features restrict different paths by which heat could enter or leave.
The passage explains complementary roles for the evacuated gap, reflective surfaces, and stopper while acknowledging remaining heat paths.

157. HSPT-R14-045 - Answer C: Radiation across the gap would be less restricted, while gas circulation there would still be prevented.
Reflective surfaces restrict radiation; keeping the evacuated gap preserves the lack of gas needed for convection there.

158. HSPT-R14-046 - Answer B: had its air removed
The preceding sentence says nearly all the air is removed, directly establishing the meaning of evacuated in this context.

159. HSPT-R14-047 - Answer D: It adds other heat paths and clarifies why the design cannot stop heat transfer completely.
The stopper and supports extend the explanation beyond the gap and qualify the design's effectiveness by identifying remaining paths.

160. HSPT-R14-048 - Answer B: Veda was to open the door.
The opening sentence states that the door should open at the third knock; its failure creates the unplanned pause.

161. HSPT-R14-049 - Answer C: Opening the door would disrupt the new interaction the actors have built around its remaining closed.
The narrator recognizes that the closed door now supports the conversation, so an immediate repair would interrupt the actors' adaptation.

162. HSPT-R14-050 - Answer A: a secure basis for continuing
The following phrase calls purchase a foothold, showing that the scene has regained support or traction for continuing.

163. HSPT-R14-051 - Answer D: They turn the accidental closed door into an apparently intentional expression of the character's guarded welcome.
The visitor's remark and Veda's reply make the physical obstacle serve the characters' interaction, allowing the scene to continue.

164. HSPT-R14-052 - Answer B: It contrasts the door's temporary role in the invented scene with its ordinary practical function after repair.
During the scene, the stuck door became a dramatic refusal; afterward, Veda playfully returns it to its practical identity.

165. HSPT-R14-053 - Answer D: Its gradually sloping face
The second paragraph explains that a sloping face meets the arm and pushes it outward against the spring.

166. HSPT-R14-054 - Answer A: Unequal tooth faces work with a positioned, spring-pressed arm to permit one rotation direction and resist the other.
The explanation coordinates tooth shape, arm position, and spring action to account for the mechanism's directional behavior.

167. HSPT-R14-055 - Answer C: The arm would no longer provide the described barrier to backward rotation.
The final sentence states that lifting the arm clear removes the stop, so the backward-moving teeth would no longer meet that barrier.

168. HSPT-R14-056 - Answer C: makes operative contact with
The arm is physically contacting the tooth to block movement, so engages means makes operative contact rather than interests or employs.

169. HSPT-R14-057 - Answer B: The contrast shows why the spring by itself cannot account for the directional effect.
Changing the face relationship while retaining a spring isolates the importance of tooth shape and arm position to one-way operation.

170. HSPT-R14-058 - Answer A: A binder supplied the manuscript's cover.
The opening note explicitly associates 1911 with the binder's cover, while the freeze and surviving copy carry different dates.

171. HSPT-R14-059 - Answer B: A date's evidentiary meaning depends on which event or stage in a document's history it identifies.
The passage distinguishes event, composition, copying, and binding so that each date supports an appropriate claim rather than every claim.

172. HSPT-R14-060 - Answer C: The account already existed before the surviving copy made in 1780.
A copy from 1750 demonstrates that the account existed by then, but does not verify its details or the later copyist's experience.

173. HSPT-R14-061 - Answer D: provides evidence of
The note supports the claim that copying occurred in 1780; it provides evidence of that act rather than commands or predicts it.

174. HSPT-R14-062 - Answer D: It exposes different dated stages within one object before explaining why they must be distinguished.
Moving inward reveals several dates associated with different acts, setting up the explanation that follows rather than arranging a single chronology.

Mathematics

175. HSPT-M14-001 - Answer C: 1
Multiplying any real number by 1 leaves it unchanged; using n = 2 already rules out the other proposed replacements.

176. HSPT-M14-002 - Answer B: 14
Squaring −14 gives 196, and the radical sign denotes the nonnegative square root, so the result is 14 rather than −14.

177. HSPT-M14-003 - Answer C: It is irrational because it never terminates or eventually repeats a fixed block.
The zero runs grow without bound while 1s keep occurring, ruling out an eventually repeating block; a rule-generated decimal need not be rational.

178. HSPT-M14-004 - Answer B: 7
The product is 720: 5, 9, and 12 divide it exactly, whereas 7 does not because no factor contributes a prime factor of 7.

179. HSPT-M14-005 - Answer A: 12
The given identity assigns the same output to opposite inputs; substituting x = 7 gives f(−7) = f(7) = 12.

180. HSPT-M14-006 - Answer D: Subtract 11, then divide by 4.
Subtracting 11 reverses the final addition; dividing by 4 then reverses the multiplication, so ((4x + 11) − 11)/4 = x.

181. HSPT-M14-007 - Answer B: Every real k except 2
When k ≠ 2, the nonzero coefficient gives one solution, x = k + 2. At k = 2, the equation becomes 0 = 0 and permits every x.

182. HSPT-M14-008 - Answer C: 4
Rewrite the expression as $(x - 3)^2 + 4$. The square cannot be negative and equals zero at x = 3, so the minimum is 4.

183. HSPT-M14-009 - Answer A: 0
Distinct parallel planes do not intersect; any common point would contradict their being separate parallel planes, so they have no points in common.

184. HSPT-M14-010 - Answer D: 2 units
With the legs on coordinate axes, the hypotenuse satisfies x/6 + y/3 = 1. The opposite square vertex is (s, s), giving s/2 = 1 and s = 2.

185. HSPT-M14-011 - Answer C: the vertical line x = 1
The midpoint is (1, 0), and all equidistant points lie on the perpendicular bisector of AB, the vertical line through that midpoint.

186. HSPT-M14-012 - Answer A: 10°C
Subtracting the two conversion equations cancels the fixed 32 offset; the change is (5/9) × 18 = 10 degrees Celsius.

187. HSPT-M14-013 - Answer A: 36 milliliters
Two minutes contains 120/20 = 6 complete cycles; each delivers 3 × 2 = 6 milliliters, for 6 × 6 = 36 milliliters.

188. HSPT-M14-014 - Answer D: a census of the target group
Because every member of the defined target group is observed, the count is a census; neither partial selection nor an assigned treatment is described.

189. HSPT-M14-015 - Answer B: 1/2, 1/4, 1/8, 1/8
T requires one specified toss, HT two, and HHT or HHH three; their probabilities are (1/2), $(1/2)^2$, $(1/2)^3$, and $(1/2)^3$.

190. HSPT-M14-016 - Answer D: 12
The replacement reverses the ordered list, but its middle observation stays in the middle; replacing that original middle value gives 20 − 8 = 12.

191. HSPT-M14-017 - Answer D: 9
Since 8! = 8 × 7!, the numerator is (8 + 1) × 7!; dividing by 7! leaves 9 without computing either full product.

192. HSPT-M14-018 - Answer A: 3 and 2
Rounding 2.46 to tenths gives 2.5, which rounds upward to 3; direct rounding of 2.46 to a whole number gives 2.

193. HSPT-M14-019 - Answer B: The first report has the smaller absolute error but the larger percentage error.
The absolute errors are 0.2 and 5 meters. Their percentage errors are (0.2/2) × 100% = 10% and (5/100) × 100% = 5%, reversing the ranking.

194. HSPT-M14-020 - Answer D: 0
A count of complete reels must be a nonnegative integer, including zero when none are inspected; the other choices are negative or nonintegral.

195. HSPT-M14-021 - Answer C: $x^2 - 3x - 10$
The required factors are (x + 2)(x − 5); expanding gives $x^2 - 3x - 10$, with the stated zeros and leading coefficient.

196. HSPT-M14-022 - Answer C: x > 2
Subtracting 1 gives 3/(x − 2) ≥ 0. Its numerator is positive, so its denominator must be positive; x = 2 is undefined.

197. HSPT-M14-023 - Answer A: −1 and 1/2
The negative branch gives x + 2 = 1, so x = −1; the nonnegative branch gives 2x = 1, so x = 1/2, and both satisfy their branch conditions.

198. HSPT-M14-024 - Answer D: right and isosceles
The right angle determines the angle classification; the equal legs determine the side classification as isosceles, so both descriptions apply together.

199. HSPT-M14-025 - Answer C: 15 square units
The common horizontal interval is 3 to 8, of length 5; the common vertical interval is 2 to 5, of length 3, giving 5 × 3 = 15.

200. HSPT-M14-026 - Answer B: 2 units
Joining the center to the vertices forms three triangles with height r; their areas total (1/2)r(9 + 10 + 17) = 18r = 36, so r = 2.

201. HSPT-M14-027 - Answer B: 9 centimeters
The thread preserves the curved line's length when straightened; its ruler span is the difference 13 − 4 = 9 centimeters, not the final reading alone.

202. HSPT-M14-028 - Answer D: 6.51 square meters
With both dimensions positive and independently allowed, area is greatest at both upper endpoints: 2.1 × 3.1 = 6.51 square meters.

203. HSPT-M14-029 - Answer A: 1.28 kilometers
Two resets account for 2,000 meters of counter progression; adding the final reading and subtracting the initial gives 2,000 + 140 − 860 = 1,280 meters = 1.28 kilometers.

204. HSPT-M14-030 - Answer C: a scatterplot with one point per meteorite, positioned by its mass and volume
A scatterplot preserves both measurements for each object as a coordinate pair, allowing their relationship to be examined rather than discarding one or both measurements.

205. HSPT-M14-031 - Answer B: 4.4
Sample A represents 4.2, 4.4, and 5.1; Sample B represents 4.4, 4.8, and 5.0, so their common measurement is 4.4.

206. HSPT-M14-032 - Answer A: 7/36
There are $4^2 = 16$ pairs with both numbers at most 4, and $3^2 = 9$ with both at most 3; the difference gives 7 of 36 pairs.

207. HSPT-M14-033 - Answer B: 1260
There are 20 terms. Pair the first with the last to make ten pairs of 126, giving 10 × 126 = 1260.

208. HSPT-M14-034 - Answer B: {−1, 1}
The reciprocals of 1 and −1 are integers. Zero has no reciprocal, and integers with greater absolute value have noninteger reciprocals.

209. HSPT-M14-035 - Answer C: 4
Square divisors require even exponents. The exponent of 2 can be 0 or 2, and that of 5 can be 0 or 2: four choices.

210. HSPT-M14-036 - Answer D: 5
The numerator is 5/6 and the denominator is 1/6. Dividing 5/6 by 1/6 gives 5.

211. HSPT-M14-037 - Answer B: 5
The nonzero terms have exponents 5, 2, and 0. The greatest exponent is 5, regardless of the coefficients.

212. HSPT-M14-038 - Answer A: y = 2x − 5
From x = t + 2, obtain t = x − 2. Substitution gives y = 2(x − 2) − 1 = 2x − 5.

213. HSPT-M14-039 - Answer C: 7
Three P and four Q use 13 bars and 11 panels. Eight stands would require at most two P by the bar limit but at least five P by the panel limit.

214. HSPT-M14-040 - Answer A: −21
Only the two constant terms can produce a term without x. Their product is −3 × 7 = −21.

215. HSPT-M14-041 - Answer C: A rectangle
Joining opposite edges of a rectangle makes one dimension the circumference and the other the height of an open cylinder.

216. HSPT-M14-042 - Answer D: 9π − 18 square units
The quarter-circle sector has area 9π. Its right triangle has area 6 × 6 ÷ 2 = 18; subtract to obtain 9π − 18.

217. HSPT-M14-043 - Answer B: 5√2 units
The perpendicular through (8, 0) is y = −x + 8. It meets the given line at (3, 5), whose distance from (8, 0) is √50 = 5√2.

218. HSPT-M14-044 - Answer D: 4%
Convert 0.45 kilometer to 450 meters. The grade is 100 × 18 ÷ 450 = 4%.

219. HSPT-M14-045 - Answer D: 5.4π grams
The inner radius is 1 − 0.2 = 0.8 cm. Plastic volume is $\pi(1^2 - 0.8^2) \times 10 = 3.6\pi$ cm³; multiply by 1.5 to obtain 5.4π grams.

220. HSPT-M14-046 - Answer A: Service ratings: poor, fair, good, excellent
The service ratings represent increasing levels of quality. Colors, identification codes, and species do not inherently rank levels.

221. HSPT-M14-047 - Answer C: 1/6
The three specified pennants have six equally likely relative orders, regardless of D and E. Exactly one order is A, then B, then C.

222. HSPT-M14-048 - Answer A: Five is the expected count, but the observed count can be different.
The expected count is 20 × 1/4 = 5, but randomness allows other counts, including no red results.

223. HSPT-M14-049 - Answer C: 0.999... and 1
The infinite decimal 0.999... equals 1, giving two representations of one number. The other pairs differ by truncation or place value.

224. HSPT-M14-050 - Answer D: x is irrational.
If x were rational, multiplying by 3 and adding 2 would remain rational. Therefore x must be irrational.

225. HSPT-M14-051 - Answer C: 6
The quotient equals 1 + 12/n, so n must divide 12. Its positive possibilities are 1, 2, 3, 4, 6, and 12.

226. HSPT-M14-052 - Answer B: $7x^2y$
The expression $7x^2y$ is one product with exponents 2 and 1. The alternatives involve a variable denominator, a sum, or a fractional power.

227. HSPT-M14-053 - Answer B: m > 4
Require 12 + 3m < 6m. Subtract 3m and divide by 3 to obtain m > 4; at 4 minutes the costs are equal.

228. HSPT-M14-054 - Answer D: 10
Adding five copies of 1 gives f(5) = 30. Adding three copies of 5/3 gives 3f(5/3) = f(5), so f(5/3) = 10.

229. HSPT-M14-055 - Answer A: No real value satisfies both inequalities.
The first inequality requires x > 4, while the second requires x ≤ 3. Those intervals have no common values.

230. HSPT-M14-056 - Answer A: All its sides are equal and all its interior angles are equal.
A regular polygon is both equilateral and equiangular; either condition alone permits polygons that are not regular.

231. HSPT-M14-057 - Answer B: 6 units
The first chord's segment product is 3 × 8 = 24. Thus 4 × PD = 24, giving PD = 6 units.

232. HSPT-M14-058 - Answer B: 8 units
Write the diagonals as 2k and 3k. Their area is $3k^2 = 48$, so k = 4 and the shorter diagonal is 2k = 8.

233. HSPT-M14-059 - Answer A: 00:07
The hour immediately after midnight is written 00 in 24-hour notation; seven minutes after midnight is therefore 00:07.

234. HSPT-M14-060 - Answer A: 16 meters
The original 1-centimeter distance becomes 2.5 centimeters while still representing 40 meters. Each new centimeter represents 40 ÷ 2.5 = 16 meters.

235. HSPT-M14-061 - Answer D: 60.6 meters
The round trip lasts 0.08 second and covers 120 meters. Half is 60 meters below the transducer; adding its 0.6-meter depth gives 60.6 meters.

236. HSPT-M14-062 - Answer C: 6
Exactly two modes require Y to tie Z at the greatest frequency, 6. A tie at 4 would remain below Z.

237. HSPT-M14-063 - Answer D: A has the higher percentage at each range; B has the higher percentage overall.
A leads 90% to 80% near and 50% to 40% far. Overall A has 39/70, about 55.7%, while B has 76/100 = 76%.

238. HSPT-M14-064 - Answer C: 1/3
Of the six assignments, only QRP and RPQ place every nameplate incorrectly on portraits P, Q, R. The probability is 2/6 = 1/3.

Language

239. HSPT-L14-001 - Answer A: The word committee contains two t's.
Under the stated style rule, the plural of the single lowercase letter t is t's; the apostrophe prevents confusion with an ordinary word.

240. HSPT-L14-002 - Answer D: An opening mark begins each paragraph; a closing mark appears only at the end of the second.
Opening marks at both paragraph beginnings identify continuing quoted speech; withholding the first paragraph's closing mark signals that the same quotation continues.

241. HSPT-L14-003 - Answer C: The sled—even without a rider—slid down the slope.
A pair of dashes marks both boundaries of the interruption; mixing a dash with a comma or omitting its partner breaks the requested enclosure.

242. HSPT-L14-004 - Answer D: We sent the invitation to Amira de la Vega.
The given personal name retains capital A and V with lowercase de la; being part of a name does not capitalize every connecting word.

243. HSPT-L14-005 - Answer B: To Whom It May Concern:
The conventional general salutation capitalizes each word in To Whom It May Concern, including the short function words It and May.

244. HSPT-L14-006 - Answer C: maneuver
Maneuver is the standard US spelling, retaining the vowel sequence eu after man and ending in ver; the alternatives transpose, substitute, or omit vowels.

245. HSPT-L14-007 - Answer C: souvenir
Souvenir begins with sou and ends with venir; the alternatives remove the initial o or add and rearrange letters in the ending.

246. HSPT-L14-008 - Answer C: which
Which can refer to the preceding event as a whole; the lock's freezing caused the delay, rather than supplying a person as an object or introducing a restrictive clause.

247. HSPT-L14-009 - Answer A: The more tightly the cord is twisted, the less easily it bends.
The linked construction uses the more ... the less with comparative forms; a superlative or a double comparative breaks that grammatical pattern.

248. HSPT-L14-010 - Answer A: I wish I could swim.
Could after wish expresses an unreal present ability; can treats the ability as actual, while the other phrases have incorrect verb forms.

249. HSPT-L14-011 - Answer D: faze
Faze means disturb or unsettle; phase refers to a stage or a planned sequence and does not express the presenter's composure.

250. HSPT-L14-012 - Answer B: credible; credulous
Credible describes something worthy of belief; credulous describes someone too ready to believe, so the evidence-based report and uncritical reader require different adjectives.

251. HSPT-L14-013 - Answer A: Place the three code entries on separate lines, each followed by its condition.
Separate code-and-condition lines let a reader locate the required entry quickly while retaining the distinctions needed to use each symbol correctly.

252. HSPT-L14-014 - Answer B: A scheduled test with a lost reading receives NR; a sample assigned only a color inspection receives NA.
NR identifies an absent required reading, while NA identifies a test outside the sample's assignment; the correct example preserves both categories.

253. HSPT-L14-015 - Answer D: Average only the recorded numerical readings; report required but unrecorded readings separately as NR.
Only actual numerical readings belong in the numerical average; separately reporting NR preserves missing required measurements without inventing zero results or relabeling them as inapplicable.

254. HSPT-L14-016 - Answer C: GUIDE: The tram has stopped.
The colon separates the complete speaker label from the speech that follows; it does not divide the label or words within the speech.

255. HSPT-L14-017 - Answer A: The awning was blue, not green.
The contrasting phrase begins with not, so the comma belongs immediately before that phrase, without separating the subject from its verb or complement.

256. HSPT-L14-018 - Answer D: The reading was 6 volts, written 6 V.
The written unit name stays lowercase while its case-sensitive symbol is uppercase; the correct version maintains that distinction in both forms.

257. HSPT-L14-019 - Answer B: The crowd listened as Chance asked for a chance to speak.
Chance is capitalized when it names the personified character; the later chance means an ordinary opportunity and remains lowercase.

258. HSPT-L14-020 - Answer C: casserole
Casserole uses a double s, a single l, and a final e; the alternatives omit or duplicate letters in that spelling.

259. HSPT-L14-021 - Answer D: miniature
Miniature includes the i before ature and has a single n; the alternatives omit that vowel, transpose the ending, or duplicate the consonant.

260. HSPT-L14-022 - Answer D: hypocrisy
Hypocrisy begins with hypo and ends in crisy, with s rather than a second c; the alternatives change those vowel and consonant sequences.

261. HSPT-L14-023 - Answer B: check
Emphatic did carries the past tense and takes the base verb check; the following verb does not receive a second tense ending.

262. HSPT-L14-024 - Answer D: is
A measurement phrase treated as one total distance takes a singular verb, so twelve kilometers is agrees with that unitary meaning.

263. HSPT-L14-025 - Answer B: The package appears to have been opened before it arrived.
After to, the perfect passive infinitive uses have been plus the past participle opened, locating the passive event earlier without using finite had.

264. HSPT-L14-026 - Answer A: flair
Flair means a natural aptitude or distinctive talent; flare names a burst of light or flame and does not express skill.

265. HSPT-L14-027 - Answer A: baited; bated
Baited means supplied with bait; bated means held back or restrained in the expression bated breath, so the two contexts require different forms.

266. HSPT-L14-028 - Answer C: One branch passes through a filter that catches dust.
The preceding sentence introduces branches; beginning with One branch continues that known information before adding the filter and its dust-catching function.

267. HSPT-L14-029 - Answer A: Replace The entry tube with The intake duct.
Reusing the established name makes the intended identity explicit; the alternatives introduce another part, reduce specificity, or falsely turn one duct into two.

268. HSPT-L14-030 - Answer B: After the splitter, one branch is filtered and exits low; the other bypasses the filter and exits near the ceiling.
The description assigns the filter and low exit to one branch and an unfiltered upper route to the other; the summary must preserve that separation.

269. HSPT-L14-031 - Answer B: a finely woven belt
An adverb ending in -ly ordinarily does not take a hyphen before the participle it modifies, so finely woven stays open.

270. HSPT-L14-032 - Answer D: Marta Dune, MD, sent the agenda.
The credential MD follows the name as a supplementary element and is set off with commas on both sides.

271. HSPT-L14-033 - Answer A: three days' travel
The plural days already ends in s, so its possessive form adds only an apostrophe: three days' travel.

272. HSPT-L14-034 - Answer A: The broadcast begins at noon Eastern Standard Time.
Eastern Standard Time is a formal time-zone name, so all three words are capitalized even within a sentence.

273. HSPT-L14-035 - Answer C: We changed trains in The Hague.
The city's proper name is The Hague, including its initial article, so both The and Hague retain their capitals.

274. HSPT-L14-036 - Answer B: handkerchief
Handkerchief retains the d in hand and uses the letter sequence chief; each alternative omits or changes part of that spelling.

275. HSPT-L14-037 - Answer C: chauffeur
Chauffeur contains au, a double f, and the ending eur; the alternatives omit a letter or alter that sequence.

276. HSPT-L14-038 - Answer C: whether
Whether introduces the embedded yes-or-no question after the preposition about; if does not serve that role in formal usage.

277. HSPT-L14-039 - Answer B: are
Even when they refers to one person, it takes are. The verb agrees with they rather than the earlier singular noun traveler.

278. HSPT-L14-040 - Answer A: Nola stopped speaking.
Stopped speaking means she ceased that activity; stopped to speak instead gives the purpose for interrupting some other activity.

279. HSPT-L14-041 - Answer C: hoard
A hoard is a stored collection kept for future use; horde and herd name groups, while heard is a verb.

280. HSPT-L14-042 - Answer D: canvas; canvass
Canvas is the fabric used for a painting surface; canvass is the verb meaning to solicit opinions or support.

281. HSPT-L14-043 - Answer A: foreword; forward
A foreword is an introductory contribution to a book, while forward indicates the direction in which the usher waves people.

282. HSPT-L14-044 - Answer D: The statue blinked; the pigeons fled.
The statue's unexpected action prompts the pigeons' flight, creating connected change; the other lines offer a state, inventory, or unrelated events.

283. HSPT-L14-045 - Answer B: The statue sneezed; pigeons flew away.
Removing the before pigeons makes room for flew away while preserving both agents and the six-word limit.

284. HSPT-L14-046 - Answer D: Genesis 2:7
The stated convention uses a colon to separate chapter 2 from verse 7, producing the reference Genesis 2:7.

285. HSPT-L14-047 - Answer C: The glaze looked blue-green.
Blue-green names one blended color and retains its hyphen after looked; a comma or semicolon would not join that compound.

286. HSPT-L14-048 - Answer A: Crossing the desert with Elena
Sentence style capitalizes the opening word Crossing and the proper name Elena while keeping the ordinary word desert lowercase.

287. HSPT-L14-049 - Answer A: The carvings are pre-Columbian.
Columbian derives from a proper name and remains capitalized; the attached prefix pre- stays lowercase within the sentence.

288. HSPT-L14-050 - Answer B: sheriff
Sheriff has one r and two final f letters; the alternatives incorrectly double r, omit an f, or do both.

289. HSPT-L14-051 - Answer A: mnemonic
Mnemonic begins with the unusual mn sequence and ends in ic; the alternatives insert, omit, or transpose letters.

290. HSPT-L14-052 - Answer B: connoisseur
Connoisseur retains double n, the oi sequence, double s, and the ending eur; each alternative changes part of that spelling.

291. HSPT-L14-053 - Answer D: seem; remains
A number of means several and takes seem; the number refers to one numerical total and takes remains.

292. HSPT-L14-054 - Answer D: Weather permitting
Weather permitting is an absolute phrase with its own subject and participle, giving a condition without introducing a separate finite clause.

293. HSPT-L14-055 - Answer C: leave
Had better is followed by the base form of the main verb, so leave is required despite the word had.

294. HSPT-L14-056 - Answer C: pore
To pore over something is to examine it closely; pour concerns liquid, poor is an adjective, and paw does not fit the phrase.

295. HSPT-L14-057 - Answer C: vial; vile
A vial is a small container, while vile describes something extremely unpleasant, such as the smell in the sentence.

296. HSPT-L14-058 - Answer B: complacent; complaisant
Complacent means excessively self-satisfied, explaining ignored warnings; complaisant means eager to accommodate others, explaining agreement to every request.

297. HSPT-L14-059 - Answer D: A wicket gate is a small gate within a larger gate that permits passage without opening the larger one.
The correct sentence identifies the class, small gate, and the distinguishing placement and function; the alternatives are vague, circular, or merely negative.

298. HSPT-L14-060 - Answer A: A beam resting on supports at both ends is not a cantilever, even when its middle has no support.
The contrasting beam lacks the required free end, making the boundary explicit without replacing the definition with examples or an incorrect broader claim.